

Social Media Sentiment Analysis

```
In [13]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from colorama import Fore, init
import plotly.express as px

import string
import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer
from nltk.sentiment import SentimentIntensityAnalyzer
from nltk import tokenize
from nltk.tokenize import sent_tokenize
from nltk.tokenize import word_tokenize
from tqdm.notebook import tqdm
from collections import Counter
from wordcloud import WordCloud

nltk.download('vader_lexicon')
nltk.download('punkt')
nltk.download('stopwords')

import warnings
warnings.filterwarnings('ignore')
```

```
[nltk_data] Downloading package vader_lexicon to
[nltk_data] C:\Users\BaddD\AppData\Roaming\nltk_data...
[nltk_data] Package vader_lexicon is already up-to-date!
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\BaddD\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\BaddD\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!
```

```
In [14]: df = pd.read_csv("dataset/sentimentdataset.csv")
```

```
In [15]: df.head()
```

Out[15]:

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hash
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nc #
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Ti #Mor
2	2	2	Just finished an amazing workout! 🍌 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fit #Wor
3	3	3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#T #Adver
4	4	4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Coc #I

In [16]:

```
def null_count():  
    return pd.DataFrame({'features': df.columns,  
                          'dtypes': df.dtypes.values,  
                          'NaN count': df.isnull().sum().values,  
                          'NaN percentage': df.isnull().sum().values/df.shape[0]}).style.background  
null_count()
```

```
Out[16]:
```

	features	dtypes	NaN count	NaN percentage
0	Unnamed: 0.1	int64	0	0.000000
1	Unnamed: 0	int64	0	0.000000
2	Text	object	0	0.000000
3	Sentiment	object	0	0.000000
4	Timestamp	object	0	0.000000
5	User	object	0	0.000000
6	Platform	object	0	0.000000
7	Hashtags	object	0	0.000000
8	Retweets	float64	0	0.000000
9	Likes	float64	0	0.000000
10	Country	object	0	0.000000
11	Year	int64	0	0.000000
12	Month	int64	0	0.000000
13	Day	int64	0	0.000000
14	Hour	int64	0	0.000000

```
In [17]: df.duplicated().sum()
```

```
Out[17]: np.int64(0)
```

```
In [18]: df.columns
```

```
Out[18]: Index(['Unnamed: 0.1', 'Unnamed: 0', 'Text', 'Sentiment', 'Timestamp', 'User',  
               'Platform', 'Hashtags', 'Retweets', 'Likes', 'Country', 'Year', 'Month',  
               'Day', 'Hour'],  
              dtype='object')
```

```
In [19]: for column in df.columns:  
          num_distinct_values = len(df[column].unique())  
          print(f"{column}: {num_distinct_values} distinct values")
```

```
Unnamed: 0.1: 732 distinct values
Unnamed: 0: 732 distinct values
Text: 707 distinct values
Sentiment: 279 distinct values
Timestamp: 683 distinct values
User: 685 distinct values
Platform: 4 distinct values
Hashtags: 697 distinct values
Retweets: 26 distinct values
Likes: 38 distinct values
Country: 115 distinct values
Year: 14 distinct values
Month: 12 distinct values
Day: 31 distinct values
Hour: 22 distinct values
```

```
In [20]: df = df.drop(columns=['Unnamed: 0.1', 'Unnamed: 0', 'Hashtags', 'Day', 'Hour', 'Senti
```

```
In [23]: df['Platform'].value_counts()
```

```
Out[23]: Platform
Instagram    258
Twitter      243
Facebook     231
Name: count, dtype: int64
```

```
In [22]: df['Platform'] = df['Platform'].str.strip()
```

```
In [26]: df['Country'].value_counts()
```

```
Out[26]: Country
USA      188
UK       143
Canada   135
Australia 75
India    70
Brazil   17
France   16
Japan    15
Germany  14
Italy    11
Spain     6
South Africa 6
Greece    5
Netherlands 4
Switzerland 3
Sweden    2
Portugal  2
Austria   2
Belgium   2
Denmark   2
Czech Republic 2
Jordan    1
Peru      1
Maldives  1
China     1
Cambodia  1
Norway    1
Colombia  1
Ireland   1
Jamaica   1
Kenya     1
Scotland  1
Thailand   1
Name: count, dtype: int64
```

```
In [25]: df['Country'] = df['Country'].str.strip()
```

```
In [30]: df['Timestamp'] = pd.to_datetime(df['Timestamp'])

df['Day_of_Week'] = df['Timestamp'].dt.day_name()
```

```
In [27]: month_mapping = {
    1: 'Januari',
    2: 'Februari',
    3: 'Maret',
    4: 'April',
    5: 'Mei',
    6: 'Juni',
    7: 'Juli',
    8: 'Agustus',
    9: 'September',
    10: 'Oktober',
    11: 'November',
    12: 'Desember'}
```

```

}

df['Month'] = df['Month'].map(month_mapping)

df['Month'] = df['Month'].astype('object')

```

```

In [28]: stemmer = PorterStemmer()
stop_words = set(stopwords.words('english'))

def clean(text):
    text = str(text).lower()
    text = re.sub('\[.*?\]', '', text)
    text = re.sub('https?://\S+|www\.\S+', '', text)
    text = re.sub(r'\s+', ' ', text.strip())
    text = re.sub('<.*?>+', '', text)
    text = re.sub('%s' % re.escape(string.punctuation), '', text)
    text = re.sub('\n', '', text)
    text = re.sub('\w*\d\w*', '', text)
    text = re.sub(r'^\x00-\x7F'+',', '', text)
    text = " ".join(text.split())
    tokens = word_tokenize(text)

    cleaned_tokens = [stemmer.stem(token) for token in tokens if token.lower() not

    cleaned_text = ' '.join(cleaned_tokens)

    return cleaned_text

df["Clean_Text"] = df["Text"].apply(clean)

```

```

In [31]: specified_columns = ['Platform', 'Country', 'Year', 'Month', 'Day_of_Week']

for col in specified_columns:
    total_unique_values = df[col].nunique()
    print(f'Total unique values for {col}: {total_unique_values}')

    top_values = df[col].value_counts()

    colors = [Fore.RED, Fore.GREEN, Fore.YELLOW, Fore.BLUE, Fore.MAGENTA, Fore.CYAN]

    for i, (value, count) in enumerate(top_values.items()):
        color = colors[i % len(colors)]
        print(f'{color}{value}: {count}{Fore.RESET}')

    print('\n' + '=' * 30 + '\n')

```

Total unique values for Platform: 3

Instagram: 258

Twitter: 243

Facebook: 231

=====

Total unique values for Country: 33

USA: 188

UK: 143

Canada: 135

Australia: 75

India: 70

Brazil: 17

France: 16

Japan: 15

Germany: 14

Italy: 11

Spain: 6

South Africa: 6

Greece: 5

Netherlands: 4

Switzerland: 3

Sweden: 2

Portugal: 2

Austria: 2

Belgium: 2

Denmark: 2

Czech Republic: 2

Jordan: 1

Peru: 1

Maldives: 1

China: 1

Cambodia: 1

Norway: 1

Colombia: 1

Ireland: 1

Jamaica: 1

Kenya: 1

Scotland: 1

Thailand: 1

=====

Total unique values for Year: 14

2023: 289

2019: 73

2020: 69

2021: 63

2022: 63

2018: 56

2017: 43

2016: 38

2015: 19

2011: 4

2013: 4

2012: 4

2014: 4

2010: 3

=====

Total unique values for Month: 12

Februari: 85

Januari: 82

Agustus: 78

September: 77

Juni: 71

Juli: 62

April: 51

November: 49

Oktober: 48

Mei: 46

Maret: 44

Desember: 39

=====

Total unique values for Day_of_Week: 7

Sunday: 119

Saturday: 115

Tuesday: 110

Friday: 108

Monday: 97

Thursday: 95

Wednesday: 88

=====

```
In [32]: df1 = df.copy()
```

```
In [33]: analyzer = SentimentIntensityAnalyzer()

df1['Vader_Score'] = df1['Clean_Text'].apply(lambda text: analyzer.polarity_scores(
df1['Sentiment'] = df1['Vader_Score'].apply(lambda score: 'positive' if score >= 0.
print(df1[['Clean_Text', 'Vader_Score', 'Sentiment']].head())
```

	Clean_Text	Vader_Score	Sentiment
0	enjoy beauti day park	0.4939	positive
1	traffic terribl morn	0.0000	neutral
2	finish amaz workout	0.0000	neutral
3	excit upcom weekend getaway	0.0000	neutral
4	tri new recip dinner tonight	0.0000	neutral

```
In [34]: colors = ['#66b3ff', '#99ff99', '#ffcc99']

explode = (0.1, 0, 0)

sentiment_counts = df1.groupby("Sentiment").size()
```

```

fig, ax = plt.subplots()

wedges, texts, autotexts = ax.pie(
    x=sentiment_counts,
    labels=sentiment_counts.index,
    autopct=lambda p: f'{p:.2f}%\n({int(p*sum(sentiment_counts)/100)}',
    wedgeprops=dict(width=0.7),
    textprops=dict(size=10, color="r"),
    pctdistance=0.7,
    colors=colors,
    explode=explode,
    shadow=True)

center_circle = plt.Circle((0, 0), 0.6, color='white', fc='white', linewidth=1.25)
fig.gca().add_artist(center_circle)

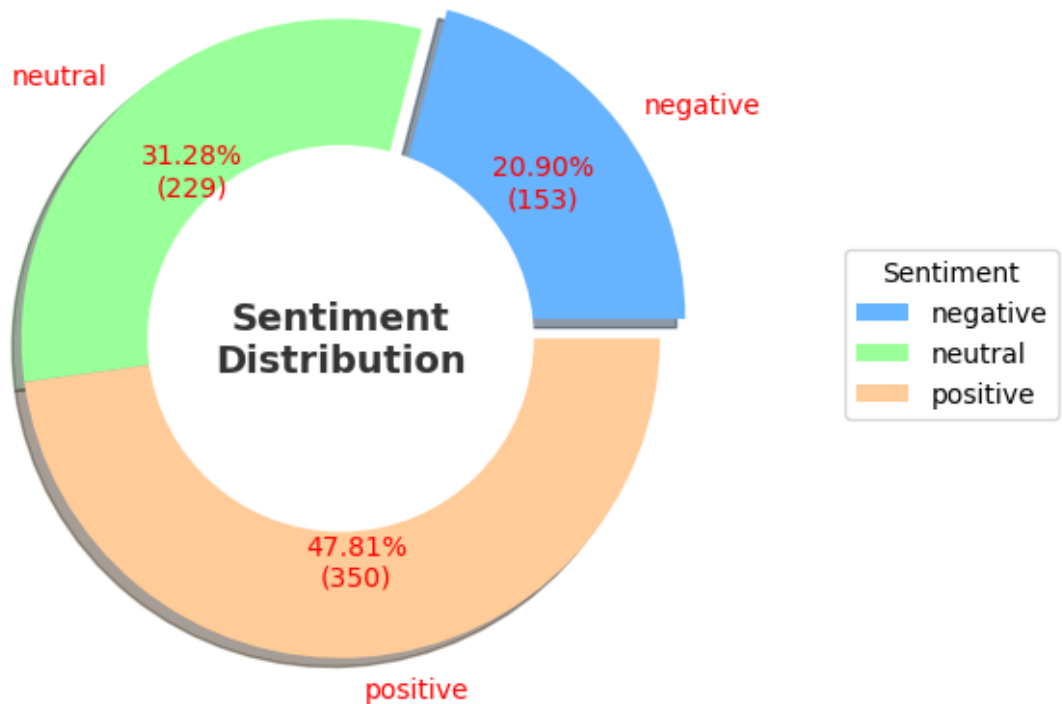
ax.text(0, 0, 'Sentiment\nDistribution', ha='center', va='center', fontsize=14, fontweight='bold')

ax.legend(sentiment_counts.index, title="Sentiment", loc="center left", bbox_to_anchor=(1, 0.5))

ax.axis('equal')

plt.show()

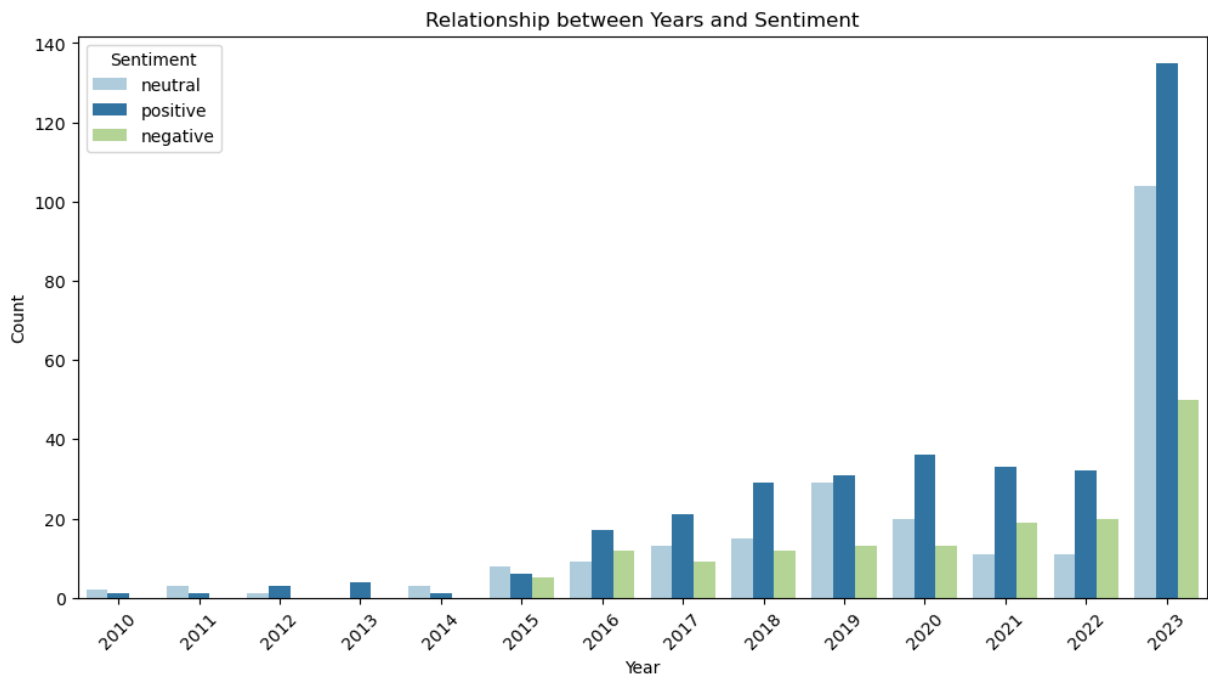
```



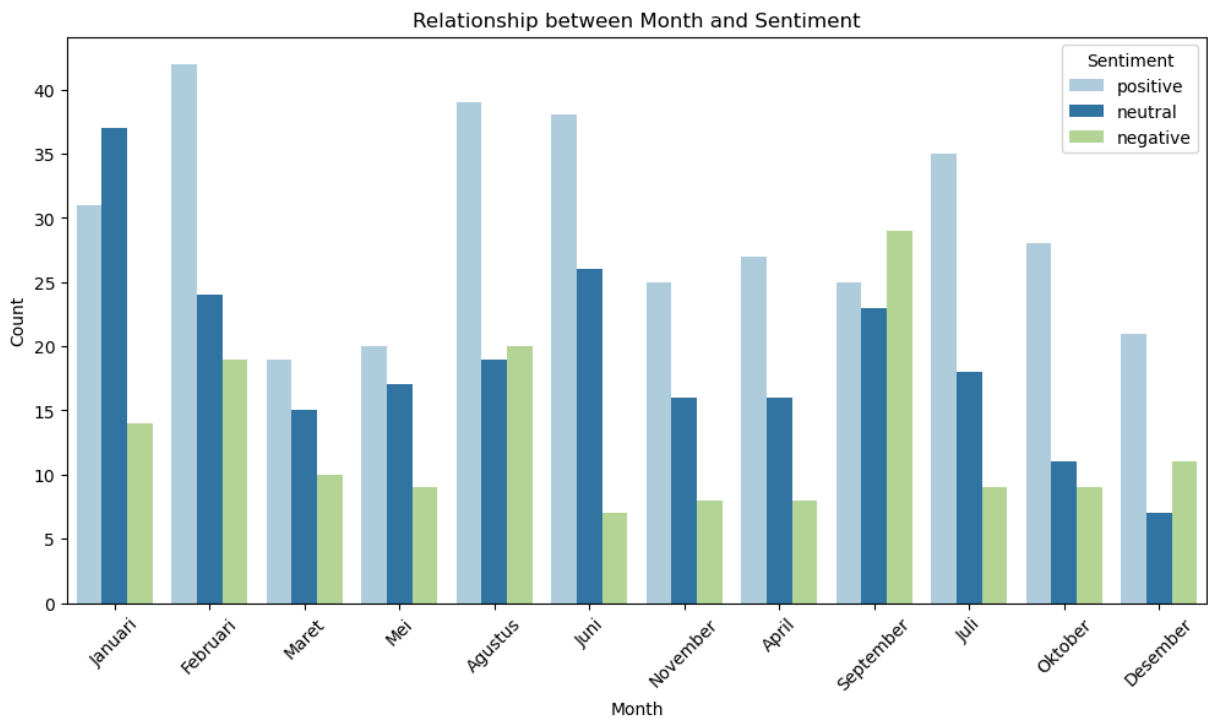
```

In [35]: plt.figure(figsize=(12, 6))
sns.countplot(x='Year', hue='Sentiment', data=df1, palette='Paired')
plt.title('Relationship between Years and Sentiment')
plt.xlabel('Year')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()

```

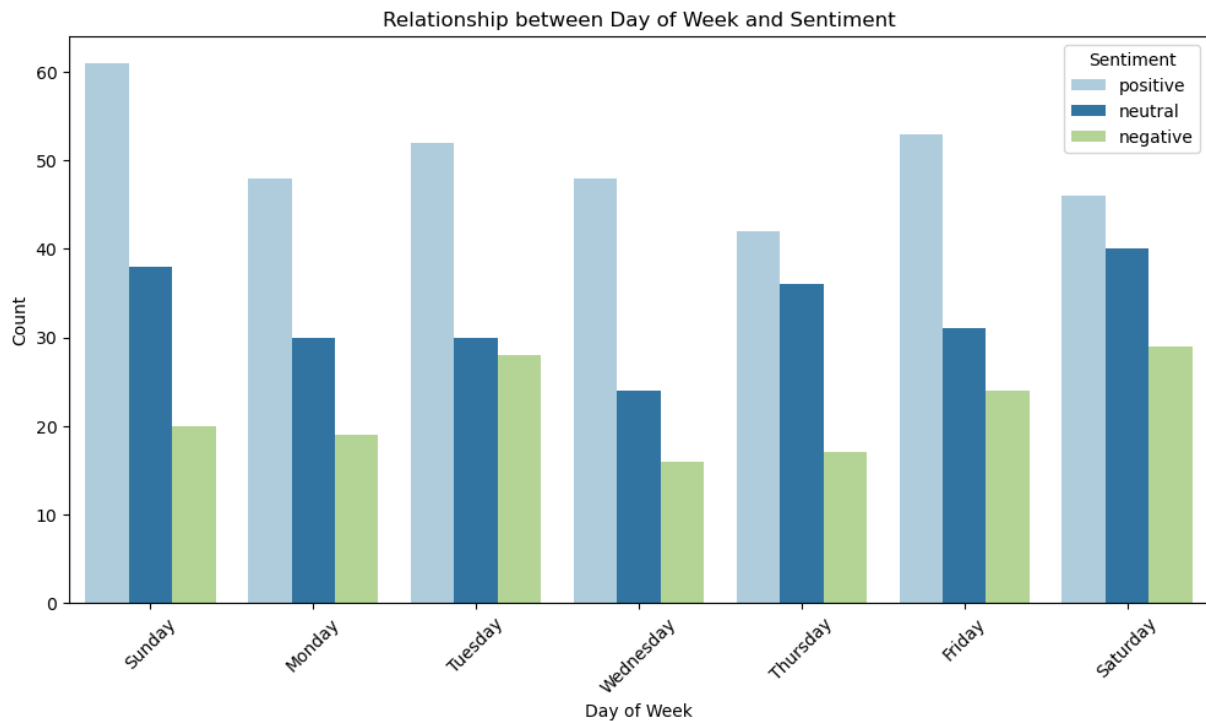


```
In [36]: plt.figure(figsize=(12, 6))
sns.countplot(x='Month', hue='Sentiment', data=df1, palette='Paired')
plt.title('Relationship between Month and Sentiment')
plt.xlabel('Month')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()
```

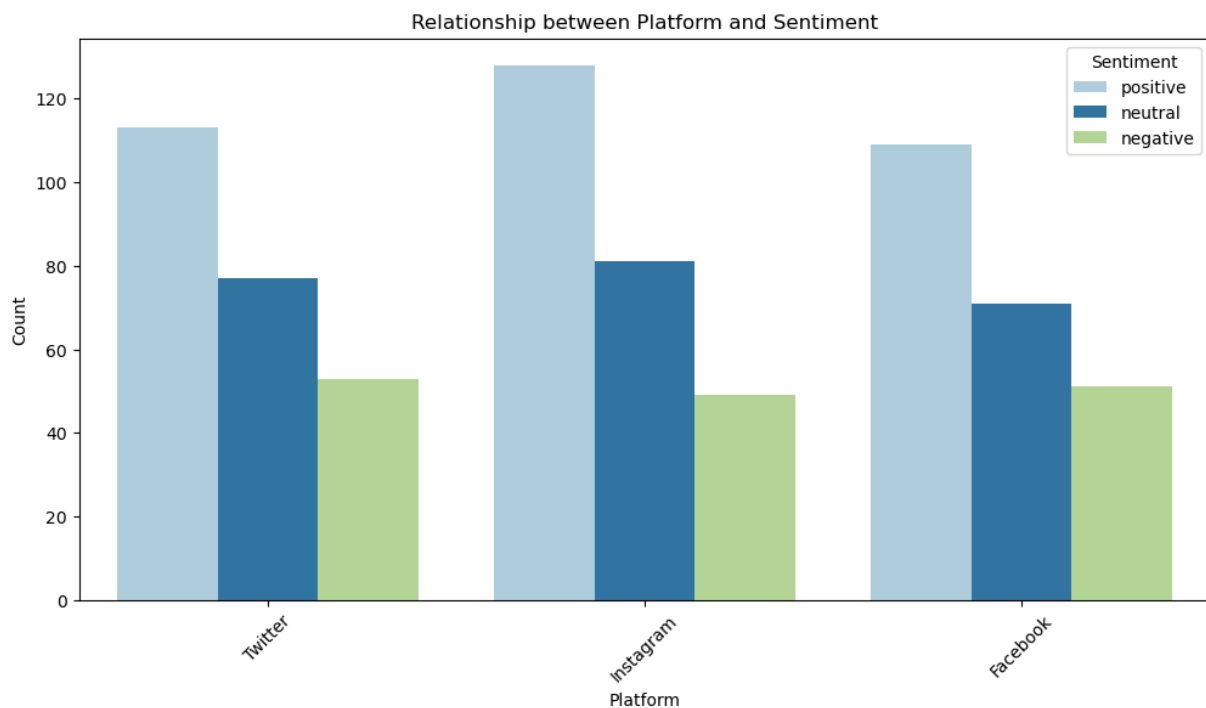


```
In [37]: plt.figure(figsize=(12, 6))
sns.countplot(x='Day_of_Week', hue='Sentiment', data=df1, palette='Paired')
plt.title('Relationship between Day of Week and Sentiment')
plt.xlabel('Day of Week')
plt.ylabel('Count')
```

```
plt.xticks(rotation=45)
plt.show()
```



```
In [38]: plt.figure(figsize=(12, 6))
sns.countplot(x='Platform', hue='Sentiment', data=df1, palette='Paired')
plt.title('Relationship between Platform and Sentiment')
plt.xlabel('Platform')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()
```



```
In [39]: plt.figure(figsize=(12, 6))
```

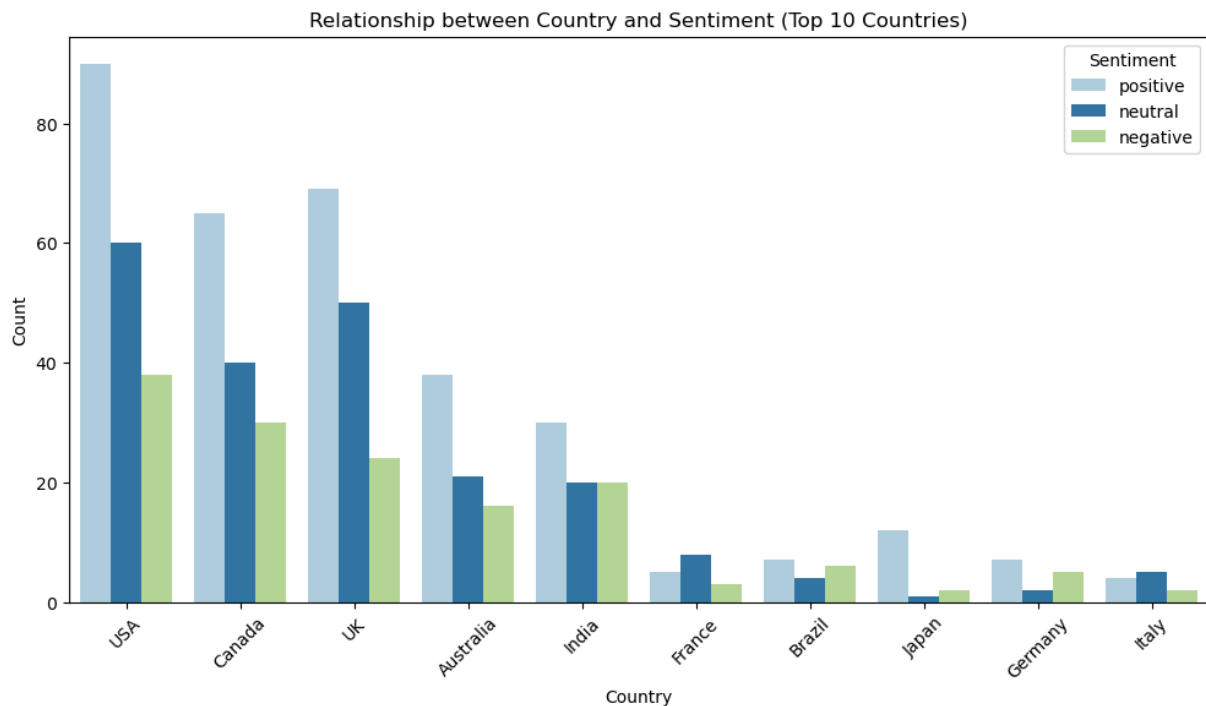
```

top_10_countries = df1['Country'].value_counts().head(10).index

df_top_10_countries = df1[df1['Country'].isin(top_10_countries)]

sns.countplot(x='Country', hue='Sentiment', data=df_top_10_countries, palette='Pair')
plt.title('Relationship between Country and Sentiment (Top 10 Countries)')
plt.xlabel('Country')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()

```



```

In [40]: df1['temp_list'] = df1['Clean_Text'].apply(lambda x: str(x).split())
top_words = Counter([item for sublist in df1['temp_list'] for item in sublist])
top_words_df = pd.DataFrame(top_words.most_common(20), columns=['Common_words', 'co

top_words_df.style.background_gradient(cmap='Blues')

```

Out[40]:

	Common_words	count
0	new	43
1	life	37
2	challeng	34
3	joy	31
4	danc	30
5	day	29
6	feel	29
7	like	28
8	dream	28
9	moment	27
10	emot	27
11	friend	26
12	heart	26
13	explor	25
14	echo	25
15	beauti	24
16	laughter	24
17	embrac	24
18	night	23
19	hope	23

```
In [41]: df1['temp_list'] = df1['Clean_Text'].apply(lambda x: str(x).split())
top_words = Counter([item for sublist in df1['temp_list'] for item in sublist])
top_words_df = pd.DataFrame(top_words.most_common(20), columns=['Common_words', 'co

fig = px.bar(top_words_df,
              x="count",
              y="Common_words",
              title='Common Words in Text Data',
              orientation='h',
              width=700,
              height=700,
              color='Common_words')

fig.show()
```

```
In [42]: Positive_sent = df1[df1['Sentiment'] == 'positive']
Negative_sent = df1[df1['Sentiment'] == 'negative']
```



```
from sklearn.svm import SVC
from sklearn.naive_bayes import MultinomialNB
from sklearn.model_selection import RandomizedSearchCV
from sklearn.metrics import confusion_matrix
```

```
In [51]: X = df2['Clean_Text'].values
        y = df2['Sentiment'].values
```

```
In [52]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

```
In [53]: vectorizer = TfidfVectorizer(max_features=5000)
        X_train_tfidf = vectorizer.fit_transform(X_train)
        X_test_tfidf = vectorizer.transform(X_test)
```

```
In [54]: pac_classifier = PassiveAggressiveClassifier(max_iter=50, random_state=42)
        pac_classifier.fit(X_train_tfidf, y_train)
```

```
Out[54]: ▼ PassiveAggressiveClassifier ⓘ ?
```

► Parameters

<u>C</u>	1.0
<u>fit_intercept</u>	True
<u>max_iter</u>	50
<u>tol</u>	0.001
<u>early_stopping</u>	False
<u>validation_fraction</u>	0.1
<u>n_iter_no_change</u>	5
<u>shuffle</u>	True
<u>verbose</u>	0
<u>loss</u>	'hinge'
<u>n_jobs</u>	None
<u>random_state</u>	42
<u>warm_start</u>	False
<u>class_weight</u>	None
<u>average</u>	False

```
In [55]: y_pred = pac_classifier.predict(X_test_tfidf)
        accuracy_test = accuracy_score(y_test, y_pred)
        classification_rep_test = classification_report(y_test, y_pred)
```

```
In [56]: print("Test Set Results:")
print(f"Accuracy: {accuracy_test}")
print("Classification Report:\n", classification_rep_test)
```

Test Set Results:

Accuracy: 0.7142857142857143

Classification Report:

	precision	recall	f1-score	support
negative	0.79	0.81	0.80	32
neutral	0.73	0.55	0.62	55
positive	0.67	0.82	0.74	60
accuracy			0.71	147
macro avg	0.73	0.72	0.72	147
weighted avg	0.72	0.71	0.71	147

```
In [57]: logistic_classifier = LogisticRegression(max_iter=50, random_state=42)
logistic_classifier.fit(X_train_tfidf, y_train)
```

Out[57]:

▼ LogisticRegression ⓘ ?	
► Parameters	
penalty	'deprecated'
C	1.0
l1_ratio	0.0
dual	False
tol	0.0001
fit_intercept	True
intercept_scaling	1
class_weight	None
random_state	42
solver	'lbfgs'
max_iter	50
verbose	0
warm_start	False
n_jobs	None

```
In [58]: y_pred_logistic = logistic_classifier.predict(X_test_tfidf)
accuracy_logistic = accuracy_score(y_test, y_pred_logistic)
classification_rep_logistic = classification_report(y_test, y_pred_logistic)
```

```
In [59]: print("Logistic Regression Results:")
print(f"Accuracy: {accuracy_logistic}")
print("Classification Report:\n", classification_rep_logistic)
```

Logistic Regression Results:

Accuracy: 0.6326530612244898

Classification Report:

	precision	recall	f1-score	support
negative	0.89	0.50	0.64	32
neutral	0.80	0.36	0.50	55
positive	0.55	0.95	0.70	60
accuracy			0.63	147
macro avg	0.75	0.60	0.61	147
weighted avg	0.72	0.63	0.61	147

```
In [60]: random_forest_classifier = RandomForestClassifier(random_state=42)
random_forest_classifier.fit(X_train_tfidf, y_train)
```

Out[60]:

▼ RandomForestClassifier ⓘ ?		
► Parameters		
<u>n_estimators</u>		100
<u>criterion</u>		'gini'
<u>max_depth</u>		None
<u>min_samples_split</u>		2
<u>min_samples_leaf</u>		1
<u>min_weight_fraction_leaf</u>		0.0
<u>max_features</u>		'sqrt'
<u>max_leaf_nodes</u>		None
<u>min_impurity_decrease</u>		0.0
<u>bootstrap</u>		True
<u>oob_score</u>		False
<u>n_jobs</u>		None
<u>random_state</u>		42
<u>verbose</u>		0
<u>warm_start</u>		False
<u>class_weight</u>		None
<u>ccp_alpha</u>		0.0
<u>max_samples</u>		None
<u>monotonic_cst</u>		None

```
In [61]: y_pred_rf = random_forest_classifier.predict(X_test_tfidf)
accuracy_rf = accuracy_score(y_test, y_pred_rf)
classification_rep_rf = classification_report(y_test, y_pred_rf)
```

```
In [62]: print("\nRandom Forest Results:")
print(f"Accuracy: {accuracy_rf}")
print("Classification Report:\n", classification_rep_rf)
```

Random Forest Results:

Accuracy: 0.6530612244897959

Classification Report:

	precision	recall	f1-score	support
negative	0.86	0.59	0.70	32
neutral	0.77	0.44	0.56	55
positive	0.56	0.88	0.69	60
accuracy			0.65	147
macro avg	0.73	0.64	0.65	147
weighted avg	0.71	0.65	0.64	147

```
In [63]: svm_classifier = SVC(random_state=42)
svm_classifier.fit(X_train_tfidf, y_train)
```

Out[63]:

▼ SVC ⓘ ?	
► Parameters	
C	1.0
kernel	'rbf'
degree	3
gamma	'scale'
coef0	0.0
shrinking	True
probability	False
tol	0.001
cache_size	200
class_weight	None
verbose	False
max_iter	-1
decision_function_shape	'ovr'
break_ties	False
random_state	42

```
In [64]: y_pred_svm = svm_classifier.predict(X_test_tfidf)
accuracy_svm = accuracy_score(y_test, y_pred_svm)
classification_rep_svm = classification_report(y_test, y_pred_svm)
```

```
In [65]: print("Support Vector Machine Results:")
```

```
print(f"Accuracy: {accuracy_svm}")
print("Classification Report:\n", classification_rep_svm)
```

Support Vector Machine Results:

Accuracy: 0.5986394557823129

Classification Report:

	precision	recall	f1-score	support
negative	1.00	0.38	0.55	32
neutral	0.94	0.31	0.47	55
positive	0.50	0.98	0.67	60
accuracy			0.60	147
macro avg	0.82	0.56	0.56	147
weighted avg	0.78	0.60	0.57	147

```
In [66]: nb_classifier = MultinomialNB()
nb_classifier.fit(X_train_tfidf, y_train)
```

```
Out[66]: ▼ MultinomialNB ⓘ ?
          ► Parameters
          ┌───┬────────┬───┐
          │   │ alpha  │ 1.0 │
          │   │ force_alpha │ True │
          │   │ fit_prior  │ True │
          │   │ class_prior │ None │
          └───┴────────┴───┘
```

```
In [67]: y_pred_nb = nb_classifier.predict(X_test_tfidf)
accuracy_nb = accuracy_score(y_test, y_pred_nb)
classification_rep_nb = classification_report(y_test, y_pred_nb)
```

```
In [68]: print("\nMultinomial Naive Bayes Results:")
print(f"Accuracy: {accuracy_nb}")
print("Classification Report:\n", classification_rep_nb)
```

Multinomial Naive Bayes Results:

Accuracy: 0.6190476190476191

Classification Report:

	precision	recall	f1-score	support
negative	1.00	0.38	0.55	32
neutral	0.90	0.35	0.50	55
positive	0.53	1.00	0.69	60
accuracy			0.62	147
macro avg	0.81	0.57	0.58	147
weighted avg	0.77	0.62	0.59	147

```
In [69]: param_dist = {
```

```
'C': [0.1, 0.5, 1.0],  
'fit_intercept': [True, False],  
'shuffle': [True, False],  
'verbose': [0, 1],  
}
```

```
In [70]: pac_classifier = PassiveAggressiveClassifier(random_state=42)  
  
randomized_search = RandomizedSearchCV(pac_classifier, param_distributions=param_di  
randomized_search.fit(X_train_tfidf, y_train)
```

```
-- Epoch 1
Norm: 1.65, NNZs: 1472, Bias: -0.360878, T: 468, Avg. loss: 0.497534, Objective: 0.497534
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.63, NNZs: 1484, Bias: -0.367336, T: 936, Avg. loss: 0.082052, Objective: 0.082052
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.69, NNZs: 1484, Bias: -0.371984, T: 1404, Avg. loss: 0.018145, Objective: 0.018145
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 0.99, NNZs: 1484, Bias: -0.369816, T: 1872, Avg. loss: 0.003613, Objective: 0.003613
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 0.99, NNZs: 1484, Bias: -0.370104, T: 2340, Avg. loss: 0.000848, Objective: 0.000848
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.45, NNZs: 1484, Bias: -0.370378, T: 2808, Avg. loss: 0.000170, Objective: 0.000170
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.24, NNZs: 1484, Bias: -0.370412, T: 3276, Avg. loss: 0.000023, Objective: 0.000023
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.24, NNZs: 1484, Bias: -0.370415, T: 3744, Avg. loss: 0.000003, Objective: 0.000003
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.24, NNZs: 1484, Bias: -0.370416, T: 4212, Avg. loss: 0.000001, Objective: 0.000001
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.24, NNZs: 1484, Bias: -0.370416, T: 4680, Avg. loss: 0.000000, Objective: 0.000000
Total training time: 0.00 seconds.
Convergence after 10 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.95, NNZs: 1527, Bias: -0.300600, T: 468, Avg. loss: 0.715257, Objective: 0.715257
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 0.94, NNZs: 1536, Bias: -0.220972, T: 936, Avg. loss: 0.193798, Objective: 0.193798
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.70, NNZs: 1537, Bias: -0.218615, T: 1404, Avg. loss: 0.062539, Objective: 0.062539
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.65, NNZs: 1537, Bias: -0.227980, T: 1872, Avg. loss: 0.024262, Objective: 0.024262
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Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.52, NNZs: 1537, Bias: -0.232196, T: 2340, Avg. loss: 0.008972, Objective:
0.008972
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.53, NNZs: 1537, Bias: -0.233053, T: 2808, Avg. loss: 0.003361, Objective:
0.003361
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.53, NNZs: 1537, Bias: -0.233033, T: 3276, Avg. loss: 0.001205, Objective:
0.001205
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.53, NNZs: 1537, Bias: -0.232969, T: 3744, Avg. loss: 0.000417, Objective:
0.000417
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.53, NNZs: 1537, Bias: -0.232961, T: 4212, Avg. loss: 0.000147, Objective:
0.000147
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.53, NNZs: 1537, Bias: -0.232972, T: 4680, Avg. loss: 0.000054, Objective:
0.000054
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.53, NNZs: 1537, Bias: -0.232981, T: 5148, Avg. loss: 0.000020, Objective:
0.000020
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.00, NNZs: 1537, Bias: -0.232985, T: 5616, Avg. loss: 0.000007, Objective:
0.000007
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.28, NNZs: 1553, Bias: -0.087083, T: 468, Avg. loss: 0.775316, Objective: 0.7
75316
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.10, NNZs: 1553, Bias: -0.084278, T: 936, Avg. loss: 0.143705, Objective: 0.1
43705
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.03, NNZs: 1553, Bias: -0.091244, T: 1404, Avg. loss: 0.050550, Objective:
0.050550
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.29, NNZs: 1553, Bias: -0.094647, T: 1872, Avg. loss: 0.017655, Objective:
0.017655
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.52, NNZs: 1553, Bias: -0.096767, T: 2340, Avg. loss: 0.006567, Objective:
0.006567
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.52, NNZs: 1553, Bias: -0.097466, T: 2808, Avg. loss: 0.002462, Objective:
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0.002462
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.58, NNZs: 1553, Bias: -0.097536, T: 3276, Avg. loss: 0.000892, Objective:
0.000892
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.58, NNZs: 1553, Bias: -0.097440, T: 3744, Avg. loss: 0.000314, Objective:
0.000314
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.96, NNZs: 1553, Bias: -0.097379, T: 4212, Avg. loss: 0.000113, Objective:
0.000113
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.96, NNZs: 1553, Bias: -0.097366, T: 4680, Avg. loss: 0.000040, Objective:
0.000040
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.96, NNZs: 1553, Bias: -0.097368, T: 5148, Avg. loss: 0.000013, Objective:
0.000013
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.96, NNZs: 1553, Bias: -0.097370, T: 5616, Avg. loss: 0.000004, Objective:
0.000004
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.74, NNZs: 1487, Bias: -0.383896, T: 468, Avg. loss: 0.515574, Objective: 0.5
15574
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.65, NNZs: 1509, Bias: -0.379719, T: 936, Avg. loss: 0.107566, Objective: 0.1
07566
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.65, NNZs: 1510, Bias: -0.388700, T: 1404, Avg. loss: 0.024776, Objective:
0.024776
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.62, NNZs: 1510, Bias: -0.385147, T: 1872, Avg. loss: 0.006816, Objective:
0.006816
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.62, NNZs: 1510, Bias: -0.384538, T: 2340, Avg. loss: 0.001820, Objective:
0.001820
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.62, NNZs: 1510, Bias: -0.384805, T: 2808, Avg. loss: 0.000524, Objective:
0.000524
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.62, NNZs: 1510, Bias: -0.384812, T: 3276, Avg. loss: 0.000148, Objective:
0.000148
Total training time: 0.00 seconds.
-- Epoch 8
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Norm: 1.62, NNZs: 1510, Bias: -0.384802, T: 3744, Avg. loss: 0.000045, Objective:
0.000045
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.62, NNZs: 1510, Bias: -0.384801, T: 4212, Avg. loss: 0.000014, Objective:
0.000014
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.62, NNZs: 1510, Bias: -0.384800, T: 4680, Avg. loss: 0.000005, Objective:
0.000005
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.62, NNZs: 1510, Bias: -0.384800, T: 5148, Avg. loss: 0.000001, Objective:
0.000001
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.15, NNZs: 1535, Bias: -0.292272, T: 468, Avg. loss: 0.725337, Objective: 0.7
25337
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.10, NNZs: 1540, Bias: -0.230971, T: 936, Avg. loss: 0.220509, Objective: 0.2
20509
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.10, NNZs: 1540, Bias: -0.220276, T: 1404, Avg. loss: 0.082367, Objective:
0.082367
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.95, NNZs: 1540, Bias: -0.226239, T: 1872, Avg. loss: 0.032693, Objective:
0.032693
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.96, NNZs: 1540, Bias: -0.228037, T: 2340, Avg. loss: 0.013232, Objective:
0.013232
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.96, NNZs: 1540, Bias: -0.227515, T: 2808, Avg. loss: 0.005744, Objective:
0.005744
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.96, NNZs: 1540, Bias: -0.227005, T: 3276, Avg. loss: 0.002657, Objective:
0.002657
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.96, NNZs: 1540, Bias: -0.226851, T: 3744, Avg. loss: 0.001296, Objective:
0.001296
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.31, NNZs: 1540, Bias: -0.226836, T: 4212, Avg. loss: 0.000632, Objective:
0.000632
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.31, NNZs: 1540, Bias: -0.226836, T: 4680, Avg. loss: 0.000311, Objective:
0.000311
Total training time: 0.00 seconds.
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-- Epoch 11
Norm: 1.31, NNZs: 1540, Bias: -0.226826, T: 5148, Avg. loss: 0.000156, Objective:
0.000156
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.31, NNZs: 1540, Bias: -0.226815, T: 5616, Avg. loss: 0.000080, Objective:
0.000080
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.31, NNZs: 1540, Bias: -0.226808, T: 6084, Avg. loss: 0.000041, Objective:
0.000041
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.35, NNZs: 1559, Bias: -0.094573, T: 468, Avg. loss: 0.794654, Objective: 0.7
94654
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.11, NNZs: 1560, Bias: -0.089989, T: 936, Avg. loss: 0.168293, Objective: 0.1
68293
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.09, NNZs: 1560, Bias: -0.098294, T: 1404, Avg. loss: 0.064311, Objective:
0.064311
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.54, NNZs: 1560, Bias: -0.100212, T: 1872, Avg. loss: 0.025444, Objective:
0.025444
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.54, NNZs: 1560, Bias: -0.102467, T: 2340, Avg. loss: 0.010539, Objective:
0.010539
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.54, NNZs: 1560, Bias: -0.104756, T: 2808, Avg. loss: 0.004808, Objective:
0.004808
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.54, NNZs: 1560, Bias: -0.105965, T: 3276, Avg. loss: 0.002222, Objective:
0.002222
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.55, NNZs: 1560, Bias: -0.106325, T: 3744, Avg. loss: 0.001021, Objective:
0.001021
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.55, NNZs: 1560, Bias: -0.106403, T: 4212, Avg. loss: 0.000467, Objective:
0.000467
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.55, NNZs: 1560, Bias: -0.106441, T: 4680, Avg. loss: 0.000217, Objective:
0.000217
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.55, NNZs: 1560, Bias: -0.106468, T: 5148, Avg. loss: 0.000103, Objective:
0.000103
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Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.55, NNZs: 1560, Bias: -0.106485, T: 5616, Avg. loss: 0.000050, Objective:
0.000050
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.55, NNZs: 1560, Bias: -0.106494, T: 6084, Avg. loss: 0.000024, Objective:
0.000024
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.86, NNZs: 1433, Bias: -0.374325, T: 468, Avg. loss: 0.493405, Objective: 0.4
93405
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.86, NNZs: 1449, Bias: -0.376590, T: 936, Avg. loss: 0.078225, Objective: 0.0
78225
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.73, NNZs: 1456, Bias: -0.381715, T: 1404, Avg. loss: 0.018507, Objective:
0.018507
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.73, NNZs: 1456, Bias: -0.380263, T: 1872, Avg. loss: 0.004390, Objective:
0.004390
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.63, NNZs: 1456, Bias: -0.379451, T: 2340, Avg. loss: 0.000803, Objective:
0.000803
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.63, NNZs: 1456, Bias: -0.379393, T: 2808, Avg. loss: 0.000134, Objective:
0.000134
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.63, NNZs: 1456, Bias: -0.379399, T: 3276, Avg. loss: 0.000025, Objective:
0.000025
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.63, NNZs: 1456, Bias: -0.379396, T: 3744, Avg. loss: 0.000005, Objective:
0.000005
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.63, NNZs: 1456, Bias: -0.379395, T: 4212, Avg. loss: 0.000001, Objective:
0.000001
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.63, NNZs: 1456, Bias: -0.379395, T: 4680, Avg. loss: 0.000000, Objective:
0.000000
Total training time: 0.00 seconds.
Convergence after 10 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.16, NNZs: 1477, Bias: -0.278074, T: 468, Avg. loss: 0.711504, Objective: 0.7
11504
Total training time: 0.00 seconds.
-- Epoch 2
```

Norm: 1.20, NNZs: 1491, Bias: -0.190892, T: 936, Avg. loss: 0.202962, Objective: 0.202962
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.19, NNZs: 1494, Bias: -0.178641, T: 1404, Avg. loss: 0.069001, Objective: 0.069001
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.67, NNZs: 1494, Bias: -0.179522, T: 1872, Avg. loss: 0.025472, Objective: 0.025472
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.69, NNZs: 1494, Bias: -0.181058, T: 2340, Avg. loss: 0.009466, Objective: 0.009466
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.53, NNZs: 1494, Bias: -0.181615, T: 2808, Avg. loss: 0.003739, Objective: 0.003739
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.54, NNZs: 1494, Bias: -0.181760, T: 3276, Avg. loss: 0.001703, Objective: 0.001703
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.54, NNZs: 1494, Bias: -0.181734, T: 3744, Avg. loss: 0.000886, Objective: 0.000886
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.55, NNZs: 1494, Bias: -0.181649, T: 4212, Avg. loss: 0.000505, Objective: 0.000505
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.55, NNZs: 1494, Bias: -0.181585, T: 4680, Avg. loss: 0.000298, Objective: 0.000298
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.55, NNZs: 1494, Bias: -0.181547, T: 5148, Avg. loss: 0.000176, Objective: 0.000176
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.55, NNZs: 1494, Bias: -0.181526, T: 5616, Avg. loss: 0.000104, Objective: 0.000104
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.32, NNZs: 1512, Bias: -0.114323, T: 468, Avg. loss: 0.767937, Objective: 0.767937
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.34, NNZs: 1513, Bias: -0.131968, T: 936, Avg. loss: 0.147810, Objective: 0.147810
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.34, NNZs: 1513, Bias: -0.145651, T: 1404, Avg. loss: 0.051966, Objective: 0.051966
Total training time: 0.00 seconds.

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-- Epoch 4
Norm: 1.02, NNZs: 1513, Bias: -0.148312, T: 1872, Avg. loss: 0.019090, Objective:
0.019090
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.01, NNZs: 1513, Bias: -0.149247, T: 2340, Avg. loss: 0.007270, Objective:
0.007270
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.01, NNZs: 1513, Bias: -0.149754, T: 2808, Avg. loss: 0.003320, Objective:
0.003320
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.01, NNZs: 1513, Bias: -0.150060, T: 3276, Avg. loss: 0.001748, Objective:
0.001748
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.01, NNZs: 1513, Bias: -0.150289, T: 3744, Avg. loss: 0.001003, Objective:
0.001003
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.01, NNZs: 1513, Bias: -0.150448, T: 4212, Avg. loss: 0.000591, Objective:
0.000591
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.01, NNZs: 1513, Bias: -0.150545, T: 4680, Avg. loss: 0.000350, Objective:
0.000350
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.01, NNZs: 1513, Bias: -0.150601, T: 5148, Avg. loss: 0.000207, Objective:
0.000207
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.01, NNZs: 1513, Bias: -0.150633, T: 5616, Avg. loss: 0.000122, Objective:
0.000122
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.65, NNZs: 1439, Bias: -0.388586, T: 468, Avg. loss: 0.504663, Objective: 0.5
04663
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.64, NNZs: 1456, Bias: -0.392100, T: 936, Avg. loss: 0.089429, Objective: 0.0
89429
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.64, NNZs: 1464, Bias: -0.399174, T: 1404, Avg. loss: 0.026085, Objective:
0.026085
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.76, NNZs: 1464, Bias: -0.397431, T: 1872, Avg. loss: 0.007479, Objective:
0.007479
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.76, NNZs: 1464, Bias: -0.397271, T: 2340, Avg. loss: 0.001878, Objective:
0.001878
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Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.76, NNZs: 1464, Bias: -0.397511, T: 2808, Avg. loss: 0.000514, Objective:
0.000514
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.76, NNZs: 1464, Bias: -0.397543, T: 3276, Avg. loss: 0.000154, Objective:
0.000154
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.76, NNZs: 1464, Bias: -0.397537, T: 3744, Avg. loss: 0.000047, Objective:
0.000047
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.76, NNZs: 1464, Bias: -0.397536, T: 4212, Avg. loss: 0.000014, Objective:
0.000014
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.76, NNZs: 1464, Bias: -0.397537, T: 4680, Avg. loss: 0.000004, Objective:
0.000004
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.76, NNZs: 1464, Bias: -0.397537, T: 5148, Avg. loss: 0.000001, Objective:
0.000001
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.07, NNZs: 1496, Bias: -0.286832, T: 468, Avg. loss: 0.713843, Objective: 0.7
13843
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.13, NNZs: 1498, Bias: -0.211031, T: 936, Avg. loss: 0.206851, Objective: 0.2
06851
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.13, NNZs: 1498, Bias: -0.195732, T: 1404, Avg. loss: 0.068777, Objective:
0.068777
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.56, NNZs: 1503, Bias: -0.197312, T: 1872, Avg. loss: 0.028502, Objective:
0.028502
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.16, NNZs: 1503, Bias: -0.203645, T: 2340, Avg. loss: 0.012469, Objective:
0.012469
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.18, NNZs: 1503, Bias: -0.207027, T: 2808, Avg. loss: 0.005493, Objective:
0.005493
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.19, NNZs: 1503, Bias: -0.207530, T: 3276, Avg. loss: 0.002353, Objective:
0.002353
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.20, NNZs: 1503, Bias: -0.207271, T: 3744, Avg. loss: 0.001053, Objective:
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0.001053
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.20, NNZs: 1503, Bias: -0.207155, T: 4212, Avg. loss: 0.000493, Objective:
0.000493
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.20, NNZs: 1503, Bias: -0.207118, T: 4680, Avg. loss: 0.000267, Objective:
0.000267
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.20, NNZs: 1503, Bias: -0.207101, T: 5148, Avg. loss: 0.000160, Objective:
0.000160
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.20, NNZs: 1503, Bias: -0.207090, T: 5616, Avg. loss: 0.000098, Objective:
0.000098
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.20, NNZs: 1503, Bias: -0.207082, T: 6084, Avg. loss: 0.000060, Objective:
0.000060
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.40, NNZs: 1517, Bias: -0.094775, T: 468, Avg. loss: 0.768477, Objective: 0.7
68477
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.14, NNZs: 1519, Bias: -0.093648, T: 936, Avg. loss: 0.161243, Objective: 0.1
61243
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.35, NNZs: 1519, Bias: -0.110893, T: 1404, Avg. loss: 0.054856, Objective:
0.054856
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.36, NNZs: 1519, Bias: -0.113742, T: 1872, Avg. loss: 0.019818, Objective:
0.019818
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.37, NNZs: 1519, Bias: -0.112401, T: 2340, Avg. loss: 0.007980, Objective:
0.007980
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.37, NNZs: 1519, Bias: -0.111519, T: 2808, Avg. loss: 0.003500, Objective:
0.003500
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.37, NNZs: 1519, Bias: -0.111484, T: 3276, Avg. loss: 0.001613, Objective:
0.001613
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.84, NNZs: 1519, Bias: -0.111641, T: 3744, Avg. loss: 0.000761, Objective:
0.000761
Total training time: 0.00 seconds.
-- Epoch 9
```

```
Norm: 1.84, NNZs: 1519, Bias: -0.111741, T: 4212, Avg. loss: 0.000382, Objective:
0.000382
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.84, NNZs: 1519, Bias: -0.111779, T: 4680, Avg. loss: 0.000215, Objective:
0.000215
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.84, NNZs: 1519, Bias: -0.111800, T: 5148, Avg. loss: 0.000130, Objective:
0.000130
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.84, NNZs: 1519, Bias: -0.111815, T: 5616, Avg. loss: 0.000079, Objective:
0.000079
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.03, NNZs: 1512, Bias: -0.436113, T: 468, Avg. loss: 0.511505, Objective: 0.5
11505
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.08, NNZs: 1530, Bias: -0.374549, T: 936, Avg. loss: 0.087601, Objective: 0.0
87601
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.08, NNZs: 1531, Bias: -0.375612, T: 1404, Avg. loss: 0.022341, Objective:
0.022341
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.76, NNZs: 1531, Bias: -0.381665, T: 1872, Avg. loss: 0.005872, Objective:
0.005872
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.76, NNZs: 1531, Bias: -0.382576, T: 2340, Avg. loss: 0.001312, Objective:
0.001312
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.76, NNZs: 1531, Bias: -0.382330, T: 2808, Avg. loss: 0.000253, Objective:
0.000253
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.76, NNZs: 1531, Bias: -0.382268, T: 3276, Avg. loss: 0.000043, Objective:
0.000043
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.76, NNZs: 1531, Bias: -0.382264, T: 3744, Avg. loss: 0.000007, Objective:
0.000007
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.76, NNZs: 1531, Bias: -0.382264, T: 4212, Avg. loss: 0.000002, Objective:
0.000002
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.76, NNZs: 1531, Bias: -0.382264, T: 4680, Avg. loss: 0.000000, Objective:
0.000000
Total training time: 0.00 seconds.
```

```
-- Epoch 11
Norm: 1.76, NNZs: 1531, Bias: -0.382264, T: 5148, Avg. loss: 0.000000, Objective:
0.000000
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.95, NNZs: 1554, Bias: -0.246298, T: 468, Avg. loss: 0.711438, Objective: 0.7
11438
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.17, NNZs: 1568, Bias: -0.196497, T: 936, Avg. loss: 0.198198, Objective: 0.1
98198
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.20, NNZs: 1573, Bias: -0.206779, T: 1404, Avg. loss: 0.079157, Objective:
0.079157
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.20, NNZs: 1573, Bias: -0.208715, T: 1872, Avg. loss: 0.034758, Objective:
0.034758
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.20, NNZs: 1573, Bias: -0.205987, T: 2340, Avg. loss: 0.014853, Objective:
0.014853
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.21, NNZs: 1573, Bias: -0.206581, T: 2808, Avg. loss: 0.006696, Objective:
0.006696
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.21, NNZs: 1573, Bias: -0.208634, T: 3276, Avg. loss: 0.003491, Objective:
0.003491
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.21, NNZs: 1573, Bias: -0.209781, T: 3744, Avg. loss: 0.001900, Objective:
0.001900
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.21, NNZs: 1573, Bias: -0.209984, T: 4212, Avg. loss: 0.001142, Objective:
0.001142
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.22, NNZs: 1573, Bias: -0.209992, T: 4680, Avg. loss: 0.000696, Objective:
0.000696
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.22, NNZs: 1573, Bias: -0.209964, T: 5148, Avg. loss: 0.000449, Objective:
0.000449
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.22, NNZs: 1573, Bias: -0.209923, T: 5616, Avg. loss: 0.000291, Objective:
0.000291
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.22, NNZs: 1573, Bias: -0.209897, T: 6084, Avg. loss: 0.000187, Objective:
0.000187
```

```
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.22, NNZs: 1591, Bias: -0.090217, T: 468, Avg. loss: 0.779357, Objective: 0.779357
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.48, NNZs: 1591, Bias: -0.102874, T: 936, Avg. loss: 0.157090, Objective: 0.157090
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.54, NNZs: 1592, Bias: -0.100430, T: 1404, Avg. loss: 0.062167, Objective: 0.062167
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.55, NNZs: 1592, Bias: -0.107885, T: 1872, Avg. loss: 0.027156, Objective: 0.027156
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.56, NNZs: 1592, Bias: -0.111883, T: 2340, Avg. loss: 0.011283, Objective: 0.011283
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.56, NNZs: 1592, Bias: -0.112039, T: 2808, Avg. loss: 0.005316, Objective: 0.005316
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.56, NNZs: 1592, Bias: -0.111279, T: 3276, Avg. loss: 0.002748, Objective: 0.002748
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.56, NNZs: 1592, Bias: -0.110911, T: 3744, Avg. loss: 0.001628, Objective: 0.001628
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.57, NNZs: 1592, Bias: -0.110959, T: 4212, Avg. loss: 0.001028, Objective: 0.001028
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.57, NNZs: 1592, Bias: -0.111081, T: 4680, Avg. loss: 0.000650, Objective: 0.000650
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.57, NNZs: 1592, Bias: -0.111170, T: 5148, Avg. loss: 0.000416, Objective: 0.000416
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.57, NNZs: 1592, Bias: -0.111228, T: 5616, Avg. loss: 0.000266, Objective: 0.000266
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.57, NNZs: 1592, Bias: -0.111262, T: 6084, Avg. loss: 0.000171, Objective: 0.000171
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
```

Norm: 0.74, NNZs: 1469, Bias: -0.390319, T: 468, Avg. loss: 0.498233, Objective: 0.498233
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.14, NNZs: 1483, Bias: -0.352526, T: 936, Avg. loss: 0.082125, Objective: 0.082125
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.47, NNZs: 1486, Bias: -0.369890, T: 1404, Avg. loss: 0.019073, Objective: 0.019073
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.19, NNZs: 1486, Bias: -0.367177, T: 1872, Avg. loss: 0.004357, Objective: 0.004357
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.20, NNZs: 1486, Bias: -0.367918, T: 2340, Avg. loss: 0.001460, Objective: 0.001460
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.87, NNZs: 1486, Bias: -0.368454, T: 2808, Avg. loss: 0.000467, Objective: 0.000467
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.28, NNZs: 1486, Bias: -0.368417, T: 3276, Avg. loss: 0.000163, Objective: 0.000163
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.94, NNZs: 1486, Bias: -0.368371, T: 3744, Avg. loss: 0.000047, Objective: 0.000047
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.48, NNZs: 1486, Bias: -0.368362, T: 4212, Avg. loss: 0.000011, Objective: 0.000011
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 3.11, NNZs: 1486, Bias: -0.368372, T: 4680, Avg. loss: 0.000004, Objective: 0.000004
Total training time: 0.00 seconds.
Convergence after 10 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.32, NNZs: 1526, Bias: -0.265522, T: 468, Avg. loss: 0.714142, Objective: 0.714142
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.41, NNZs: 1531, Bias: -0.191339, T: 936, Avg. loss: 0.196124, Objective: 0.196124
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.21, NNZs: 1531, Bias: -0.227724, T: 1404, Avg. loss: 0.063998, Objective: 0.063998
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.64, NNZs: 1531, Bias: -0.225231, T: 1872, Avg. loss: 0.025743, Objective: 0.025743
Total training time: 0.00 seconds.

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-- Epoch 5
Norm: 2.55, NNZs: 1531, Bias: -0.231990, T: 2340, Avg. loss: 0.012568, Objective:
0.012568
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.10, NNZs: 1531, Bias: -0.232227, T: 2808, Avg. loss: 0.005214, Objective:
0.005214
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.26, NNZs: 1531, Bias: -0.232811, T: 3276, Avg. loss: 0.002678, Objective:
0.002678
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.37, NNZs: 1531, Bias: -0.232675, T: 3744, Avg. loss: 0.001144, Objective:
0.001144
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.43, NNZs: 1531, Bias: -0.233001, T: 4212, Avg. loss: 0.000628, Objective:
0.000628
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 3.15, NNZs: 1531, Bias: -0.232633, T: 4680, Avg. loss: 0.000271, Objective:
0.000271
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.77, NNZs: 1531, Bias: -0.232985, T: 5148, Avg. loss: 0.000205, Objective:
0.000205
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.24, NNZs: 1531, Bias: -0.233068, T: 5616, Avg. loss: 0.000100, Objective:
0.000100
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.43, NNZs: 1531, Bias: -0.233001, T: 6084, Avg. loss: 0.000045, Objective:
0.000045
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.68, NNZs: 1550, Bias: -0.153067, T: 468, Avg. loss: 0.772635, Objective: 0.7
72635
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.67, NNZs: 1550, Bias: -0.070661, T: 936, Avg. loss: 0.149661, Objective: 0.1
49661
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.86, NNZs: 1551, Bias: -0.107904, T: 1404, Avg. loss: 0.054768, Objective:
0.054768
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.55, NNZs: 1551, Bias: -0.101373, T: 1872, Avg. loss: 0.021681, Objective:
0.021681
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.22, NNZs: 1552, Bias: -0.105904, T: 2340, Avg. loss: 0.009829, Objective:
0.009829
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Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.00, NNZs: 1552, Bias: -0.104972, T: 2808, Avg. loss: 0.004051, Objective:
0.004051
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.11, NNZs: 1552, Bias: -0.103912, T: 3276, Avg. loss: 0.002133, Objective:
0.002133
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.64, NNZs: 1552, Bias: -0.104010, T: 3744, Avg. loss: 0.001006, Objective:
0.001006
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.12, NNZs: 1552, Bias: -0.104526, T: 4212, Avg. loss: 0.000439, Objective:
0.000439
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 0.94, NNZs: 1552, Bias: -0.104218, T: 4680, Avg. loss: 0.000258, Objective:
0.000258
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 0.99, NNZs: 1552, Bias: -0.104279, T: 5148, Avg. loss: 0.000098, Objective:
0.000098
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.60, NNZs: 1552, Bias: -0.104295, T: 5616, Avg. loss: 0.000057, Objective:
0.000057
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.95, NNZs: 1552, Bias: -0.104277, T: 6084, Avg. loss: 0.000030, Objective:
0.000030
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.69, NNZs: 1492, Bias: -0.423668, T: 468, Avg. loss: 0.517627, Objective: 0.5
17627
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.89, NNZs: 1508, Bias: -0.353326, T: 936, Avg. loss: 0.100652, Objective: 0.1
00652
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.63, NNZs: 1510, Bias: -0.392517, T: 1404, Avg. loss: 0.025578, Objective:
0.025578
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.79, NNZs: 1510, Bias: -0.374984, T: 1872, Avg. loss: 0.007982, Objective:
0.007982
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.66, NNZs: 1510, Bias: -0.386149, T: 2340, Avg. loss: 0.003947, Objective:
0.003947
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.76, NNZs: 1510, Bias: -0.382711, T: 2808, Avg. loss: 0.001306, Objective:
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0.001306
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.12, NNZs: 1510, Bias: -0.385276, T: 3276, Avg. loss: 0.000789, Objective:
0.000789
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 3.02, NNZs: 1510, Bias: -0.384774, T: 3744, Avg. loss: 0.000288, Objective:
0.000288
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.13, NNZs: 1510, Bias: -0.384892, T: 4212, Avg. loss: 0.000110, Objective:
0.000110
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.48, NNZs: 1510, Bias: -0.384954, T: 4680, Avg. loss: 0.000052, Objective:
0.000052
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.18, NNZs: 1510, Bias: -0.384924, T: 5148, Avg. loss: 0.000017, Objective:
0.000017
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.54, NNZs: 1516, Bias: -0.277238, T: 468, Avg. loss: 0.728120, Objective: 0.7
28120
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.53, NNZs: 1525, Bias: -0.204313, T: 936, Avg. loss: 0.211858, Objective: 0.2
11858
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.39, NNZs: 1525, Bias: -0.233288, T: 1404, Avg. loss: 0.079336, Objective:
0.079336
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.51, NNZs: 1525, Bias: -0.233054, T: 1872, Avg. loss: 0.035263, Objective:
0.035263
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.36, NNZs: 1525, Bias: -0.231660, T: 2340, Avg. loss: 0.016305, Objective:
0.016305
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.09, NNZs: 1525, Bias: -0.230708, T: 2808, Avg. loss: 0.008079, Objective:
0.008079
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.41, NNZs: 1525, Bias: -0.232996, T: 3276, Avg. loss: 0.004759, Objective:
0.004759
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.95, NNZs: 1525, Bias: -0.233501, T: 3744, Avg. loss: 0.002360, Objective:
0.002360
Total training time: 0.00 seconds.
-- Epoch 9
```

Norm: 2.62, NNZs: 1525, Bias: -0.231526, T: 4212, Avg. loss: 0.001128, Objective: 0.001128
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.53, NNZs: 1525, Bias: -0.231932, T: 4680, Avg. loss: 0.000874, Objective: 0.000874
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.07, NNZs: 1525, Bias: -0.232821, T: 5148, Avg. loss: 0.000434, Objective: 0.000434
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.71, NNZs: 1525, Bias: -0.232663, T: 5616, Avg. loss: 0.000300, Objective: 0.000300
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.95, NNZs: 1525, Bias: -0.232418, T: 6084, Avg. loss: 0.000139, Objective: 0.000139
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.44, NNZs: 1525, Bias: -0.232494, T: 6552, Avg. loss: 0.000117, Objective: 0.000117
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.61, NNZs: 1551, Bias: -0.095375, T: 468, Avg. loss: 0.790210, Objective: 0.790210
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.30, NNZs: 1553, Bias: -0.070739, T: 936, Avg. loss: 0.169378, Objective: 0.169378
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 0.96, NNZs: 1554, Bias: -0.092976, T: 1404, Avg. loss: 0.062335, Objective: 0.062335
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.91, NNZs: 1554, Bias: -0.086082, T: 1872, Avg. loss: 0.027100, Objective: 0.027100
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.57, NNZs: 1554, Bias: -0.094163, T: 2340, Avg. loss: 0.011819, Objective: 0.011819
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.13, NNZs: 1554, Bias: -0.093509, T: 2808, Avg. loss: 0.005009, Objective: 0.005009
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.90, NNZs: 1554, Bias: -0.092822, T: 3276, Avg. loss: 0.002901, Objective: 0.002901
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.67, NNZs: 1554, Bias: -0.093347, T: 3744, Avg. loss: 0.001553, Objective: 0.001553
Total training time: 0.00 seconds.

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-- Epoch 9
Norm: 3.81, NNZs: 1554, Bias: -0.094337, T: 4212, Avg. loss: 0.000787, Objective:
0.000787
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.86, NNZs: 1554, Bias: -0.094015, T: 4680, Avg. loss: 0.000576, Objective:
0.000576
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.44, NNZs: 1554, Bias: -0.094149, T: 5148, Avg. loss: 0.000324, Objective:
0.000324
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.00, NNZs: 1554, Bias: -0.093858, T: 5616, Avg. loss: 0.000129, Objective:
0.000129
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.53, NNZs: 1554, Bias: -0.094079, T: 6084, Avg. loss: 0.000078, Objective:
0.000078
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.83, NNZs: 1457, Bias: -0.390794, T: 468, Avg. loss: 0.491044, Objective: 0.4
91044
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.42, NNZs: 1470, Bias: -0.355903, T: 936, Avg. loss: 0.077510, Objective: 0.0
77510
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.72, NNZs: 1473, Bias: -0.372918, T: 1404, Avg. loss: 0.017517, Objective:
0.017517
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 0.88, NNZs: 1473, Bias: -0.366141, T: 1872, Avg. loss: 0.004287, Objective:
0.004287
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.73, NNZs: 1473, Bias: -0.372108, T: 2340, Avg. loss: 0.002306, Objective:
0.002306
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.70, NNZs: 1473, Bias: -0.370345, T: 2808, Avg. loss: 0.000575, Objective:
0.000575
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 0.88, NNZs: 1473, Bias: -0.371744, T: 3276, Avg. loss: 0.000404, Objective:
0.000404
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.74, NNZs: 1473, Bias: -0.371541, T: 3744, Avg. loss: 0.000157, Objective:
0.000157
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.72, NNZs: 1473, Bias: -0.371568, T: 4212, Avg. loss: 0.000050, Objective:
0.000050
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Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.46, NNZs: 1473, Bias: -0.371620, T: 4680, Avg. loss: 0.000028, Objective:
0.000028
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.72, NNZs: 1473, Bias: -0.371589, T: 5148, Avg. loss: 0.000008, Objective:
0.000008
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.52, NNZs: 1494, Bias: -0.247380, T: 468, Avg. loss: 0.713256, Objective: 0.7
13256
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.66, NNZs: 1501, Bias: -0.197606, T: 936, Avg. loss: 0.199805, Objective: 0.1
99805
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.60, NNZs: 1502, Bias: -0.183931, T: 1404, Avg. loss: 0.064491, Objective:
0.064491
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.71, NNZs: 1502, Bias: -0.184880, T: 1872, Avg. loss: 0.030633, Objective:
0.030633
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.32, NNZs: 1502, Bias: -0.185265, T: 2340, Avg. loss: 0.014086, Objective:
0.014086
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.52, NNZs: 1502, Bias: -0.184593, T: 2808, Avg. loss: 0.006857, Objective:
0.006857
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.71, NNZs: 1502, Bias: -0.185327, T: 3276, Avg. loss: 0.003838, Objective:
0.003838
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.65, NNZs: 1502, Bias: -0.185713, T: 3744, Avg. loss: 0.002233, Objective:
0.002233
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.80, NNZs: 1502, Bias: -0.183900, T: 4212, Avg. loss: 0.000877, Objective:
0.000877
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.51, NNZs: 1502, Bias: -0.184305, T: 4680, Avg. loss: 0.000743, Objective:
0.000743
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.05, NNZs: 1502, Bias: -0.185402, T: 5148, Avg. loss: 0.000404, Objective:
0.000404
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.21, NNZs: 1502, Bias: -0.185267, T: 5616, Avg. loss: 0.000323, Objective:
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0.000323
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.53, NNZs: 1502, Bias: -0.185034, T: 6084, Avg. loss: 0.000200, Objective:
0.000200
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.24, NNZs: 1502, Bias: -0.184930, T: 6552, Avg. loss: 0.000141, Objective:
0.000141
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.30, NNZs: 1511, Bias: -0.103306, T: 468, Avg. loss: 0.760159, Objective: 0.7
60159
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.61, NNZs: 1512, Bias: -0.115677, T: 936, Avg. loss: 0.146473, Objective: 0.1
46473
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.90, NNZs: 1513, Bias: -0.137742, T: 1404, Avg. loss: 0.049552, Objective:
0.049552
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 3.37, NNZs: 1513, Bias: -0.129350, T: 1872, Avg. loss: 0.021528, Objective:
0.021528
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.26, NNZs: 1513, Bias: -0.136528, T: 2340, Avg. loss: 0.009293, Objective:
0.009293
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.64, NNZs: 1513, Bias: -0.137127, T: 2808, Avg. loss: 0.004694, Objective:
0.004694
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.05, NNZs: 1513, Bias: -0.135907, T: 3276, Avg. loss: 0.002136, Objective:
0.002136
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.62, NNZs: 1513, Bias: -0.136101, T: 3744, Avg. loss: 0.001270, Objective:
0.001270
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.84, NNZs: 1513, Bias: -0.136993, T: 4212, Avg. loss: 0.000711, Objective:
0.000711
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.99, NNZs: 1513, Bias: -0.136385, T: 4680, Avg. loss: 0.000591, Objective:
0.000591
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.16, NNZs: 1513, Bias: -0.136106, T: 5148, Avg. loss: 0.000432, Objective:
0.000432
Total training time: 0.00 seconds.
-- Epoch 12
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Norm: 3.06, NNZs: 1513, Bias: -0.135876, T: 5616, Avg. loss: 0.000202, Objective:
0.000202
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.38, NNZs: 1452, Bias: -0.398522, T: 468, Avg. loss: 0.505690, Objective: 0.5
05690
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.67, NNZs: 1463, Bias: -0.387001, T: 936, Avg. loss: 0.090440, Objective: 0.0
90440
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.11, NNZs: 1463, Bias: -0.394865, T: 1404, Avg. loss: 0.024672, Objective:
0.024672
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.31, NNZs: 1463, Bias: -0.390946, T: 1872, Avg. loss: 0.007587, Objective:
0.007587
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.02, NNZs: 1463, Bias: -0.395567, T: 2340, Avg. loss: 0.003331, Objective:
0.003331
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.43, NNZs: 1463, Bias: -0.393132, T: 2808, Avg. loss: 0.001179, Objective:
0.001179
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.14, NNZs: 1463, Bias: -0.395004, T: 3276, Avg. loss: 0.000611, Objective:
0.000611
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 0.78, NNZs: 1463, Bias: -0.394660, T: 3744, Avg. loss: 0.000282, Objective:
0.000282
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.03, NNZs: 1463, Bias: -0.394624, T: 4212, Avg. loss: 0.000102, Objective:
0.000102
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.26, NNZs: 1463, Bias: -0.394759, T: 4680, Avg. loss: 0.000060, Objective:
0.000060
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.30, NNZs: 1463, Bias: -0.394703, T: 5148, Avg. loss: 0.000019, Objective:
0.000019
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.04, NNZs: 1498, Bias: -0.299330, T: 468, Avg. loss: 0.719488, Objective: 0.7
19488
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.77, NNZs: 1500, Bias: -0.211550, T: 936, Avg. loss: 0.194570, Objective: 0.1
94570
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Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.87, NNZs: 1501, Bias: -0.192091, T: 1404, Avg. loss: 0.075087, Objective:
0.075087
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.63, NNZs: 1501, Bias: -0.209930, T: 1872, Avg. loss: 0.034134, Objective:
0.034134
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.30, NNZs: 1501, Bias: -0.214819, T: 2340, Avg. loss: 0.012916, Objective:
0.012916
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.92, NNZs: 1501, Bias: -0.213905, T: 2808, Avg. loss: 0.007239, Objective:
0.007239
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.86, NNZs: 1501, Bias: -0.217965, T: 3276, Avg. loss: 0.003682, Objective:
0.003682
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 3.06, NNZs: 1501, Bias: -0.217471, T: 3744, Avg. loss: 0.002418, Objective:
0.002418
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.17, NNZs: 1501, Bias: -0.215447, T: 4212, Avg. loss: 0.001111, Objective:
0.001111
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.33, NNZs: 1501, Bias: -0.216264, T: 4680, Avg. loss: 0.000961, Objective:
0.000961
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 0.91, NNZs: 1501, Bias: -0.217259, T: 5148, Avg. loss: 0.000424, Objective:
0.000424
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.17, NNZs: 1501, Bias: -0.217109, T: 5616, Avg. loss: 0.000382, Objective:
0.000382
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.29, NNZs: 1501, Bias: -0.216853, T: 6084, Avg. loss: 0.000214, Objective:
0.000214
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.37, NNZs: 1501, Bias: -0.216868, T: 6552, Avg. loss: 0.000174, Objective:
0.000174
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.73, NNZs: 1519, Bias: -0.075806, T: 468, Avg. loss: 0.769476, Objective: 0.7
69476
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.17, NNZs: 1519, Bias: -0.097819, T: 936, Avg. loss: 0.153515, Objective: 0.1
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53515
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.78, NNZs: 1519, Bias: -0.098017, T: 1404, Avg. loss: 0.055861, Objective:
0.055861
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.44, NNZs: 1519, Bias: -0.088899, T: 1872, Avg. loss: 0.021577, Objective:
0.021577
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.80, NNZs: 1519, Bias: -0.095977, T: 2340, Avg. loss: 0.011223, Objective:
0.011223
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.42, NNZs: 1519, Bias: -0.094288, T: 2808, Avg. loss: 0.005063, Objective:
0.005063
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.42, NNZs: 1519, Bias: -0.095011, T: 3276, Avg. loss: 0.002745, Objective:
0.002745
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.49, NNZs: 1519, Bias: -0.095229, T: 3744, Avg. loss: 0.001571, Objective:
0.001571
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.61, NNZs: 1519, Bias: -0.096238, T: 4212, Avg. loss: 0.000734, Objective:
0.000734
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.75, NNZs: 1519, Bias: -0.095487, T: 4680, Avg. loss: 0.000594, Objective:
0.000594
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.11, NNZs: 1519, Bias: -0.095381, T: 5148, Avg. loss: 0.000344, Objective:
0.000344
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.07, NNZs: 1519, Bias: -0.095201, T: 5616, Avg. loss: 0.000152, Objective:
0.000152
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.16, NNZs: 1519, Bias: -0.095495, T: 6084, Avg. loss: 0.000095, Objective:
0.000095
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.33, NNZs: 1527, Bias: -0.375618, T: 468, Avg. loss: 0.512577, Objective: 0.5
12577
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.41, NNZs: 1532, Bias: -0.364790, T: 936, Avg. loss: 0.090941, Objective: 0.0
90941
Total training time: 0.00 seconds.
-- Epoch 3
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Norm: 1.54, NNZs: 1534, Bias: -0.376831, T: 1404, Avg. loss: 0.021166, Objective:
0.021166
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.01, NNZs: 1534, Bias: -0.372679, T: 1872, Avg. loss: 0.007208, Objective:
0.007208
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.95, NNZs: 1534, Bias: -0.376080, T: 2340, Avg. loss: 0.002240, Objective:
0.002240
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.49, NNZs: 1534, Bias: -0.374655, T: 2808, Avg. loss: 0.000881, Objective:
0.000881
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.46, NNZs: 1534, Bias: -0.375582, T: 3276, Avg. loss: 0.000357, Objective:
0.000357
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.70, NNZs: 1534, Bias: -0.375419, T: 3744, Avg. loss: 0.000181, Objective:
0.000181
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.22, NNZs: 1534, Bias: -0.375292, T: 4212, Avg. loss: 0.000047, Objective:
0.000047
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.06, NNZs: 1534, Bias: -0.375382, T: 4680, Avg. loss: 0.000028, Objective:
0.000028
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.10, NNZs: 1534, Bias: -0.375359, T: 5148, Avg. loss: 0.000008, Objective:
0.000008
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.37, NNZs: 1575, Bias: -0.295985, T: 468, Avg. loss: 0.710548, Objective: 0.7
10548
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.98, NNZs: 1581, Bias: -0.216396, T: 936, Avg. loss: 0.191515, Objective: 0.1
91515
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.61, NNZs: 1582, Bias: -0.208297, T: 1404, Avg. loss: 0.071496, Objective:
0.071496
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.99, NNZs: 1582, Bias: -0.216435, T: 1872, Avg. loss: 0.032887, Objective:
0.032887
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.34, NNZs: 1582, Bias: -0.218042, T: 2340, Avg. loss: 0.016354, Objective:
0.016354
Total training time: 0.00 seconds.
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-- Epoch 6
Norm: 1.72, NNZs: 1582, Bias: -0.219272, T: 2808, Avg. loss: 0.010562, Objective:
0.010562
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.73, NNZs: 1582, Bias: -0.222903, T: 3276, Avg. loss: 0.005210, Objective:
0.005210
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.25, NNZs: 1582, Bias: -0.221989, T: 3744, Avg. loss: 0.003224, Objective:
0.003224
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.29, NNZs: 1582, Bias: -0.220082, T: 4212, Avg. loss: 0.001428, Objective:
0.001428
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.35, NNZs: 1582, Bias: -0.220349, T: 4680, Avg. loss: 0.001261, Objective:
0.001261
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.73, NNZs: 1582, Bias: -0.221962, T: 5148, Avg. loss: 0.000645, Objective:
0.000645
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.04, NNZs: 1582, Bias: -0.221617, T: 5616, Avg. loss: 0.000543, Objective:
0.000543
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.98, NNZs: 1582, Bias: -0.221268, T: 6084, Avg. loss: 0.000290, Objective:
0.000290
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.54, NNZs: 1582, Bias: -0.221209, T: 6552, Avg. loss: 0.000236, Objective:
0.000236
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.97, NNZs: 1585, Bias: -0.075123, T: 468, Avg. loss: 0.775356, Objective: 0.7
75356
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.66, NNZs: 1589, Bias: -0.115264, T: 936, Avg. loss: 0.161401, Objective: 0.1
61401
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.31, NNZs: 1589, Bias: -0.098966, T: 1404, Avg. loss: 0.058711, Objective:
0.058711
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.24, NNZs: 1591, Bias: -0.090319, T: 1872, Avg. loss: 0.028545, Objective:
0.028545
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.66, NNZs: 1591, Bias: -0.098929, T: 2340, Avg. loss: 0.014426, Objective:
0.014426
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Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.16, NNZs: 1591, Bias: -0.095986, T: 2808, Avg. loss: 0.006882, Objective:
0.006882
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.95, NNZs: 1591, Bias: -0.095006, T: 3276, Avg. loss: 0.003725, Objective:
0.003725
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.08, NNZs: 1591, Bias: -0.095809, T: 3744, Avg. loss: 0.002431, Objective:
0.002431
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.45, NNZs: 1591, Bias: -0.096335, T: 4212, Avg. loss: 0.001506, Objective:
0.001506
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.55, NNZs: 1591, Bias: -0.096481, T: 4680, Avg. loss: 0.001233, Objective:
0.001233
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.80, NNZs: 1591, Bias: -0.095805, T: 5148, Avg. loss: 0.000752, Objective:
0.000752
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.03, NNZs: 1591, Bias: -0.095798, T: 5616, Avg. loss: 0.000452, Objective:
0.000452
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.56, NNZs: 1591, Bias: -0.096255, T: 6084, Avg. loss: 0.000256, Objective:
0.000256
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.16, NNZs: 1507, Bias: 0.000000, T: 468, Avg. loss: 0.561863, Objective: 0.56
1863
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.12, NNZs: 1508, Bias: 0.000000, T: 936, Avg. loss: 0.065525, Objective: 0.06
5525
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.24, NNZs: 1508, Bias: 0.000000, T: 1404, Avg. loss: 0.016479, Objective: 0.0
16479
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.41, NNZs: 1508, Bias: 0.000000, T: 1872, Avg. loss: 0.004067, Objective: 0.0
04067
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.47, NNZs: 1508, Bias: 0.000000, T: 2340, Avg. loss: 0.001209, Objective: 0.0
01209
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.94, NNZs: 1508, Bias: 0.000000, T: 2808, Avg. loss: 0.000371, Objective: 0.0
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00371
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.47, NNZs: 1508, Bias: 0.000000, T: 3276, Avg. loss: 0.000129, Objective: 0.000129
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.10, NNZs: 1508, Bias: 0.000000, T: 3744, Avg. loss: 0.000040, Objective: 0.000040
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.32, NNZs: 1508, Bias: 0.000000, T: 4212, Avg. loss: 0.000010, Objective: 0.000010
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.12, NNZs: 1508, Bias: 0.000000, T: 4680, Avg. loss: 0.000004, Objective: 0.000004
Total training time: 0.00 seconds.
Convergence after 10 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.05, NNZs: 1535, Bias: 0.000000, T: 468, Avg. loss: 0.748456, Objective: 0.748456
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.20, NNZs: 1535, Bias: 0.000000, T: 936, Avg. loss: 0.166319, Objective: 0.166319
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.43, NNZs: 1537, Bias: 0.000000, T: 1404, Avg. loss: 0.056108, Objective: 0.056108
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.76, NNZs: 1537, Bias: 0.000000, T: 1872, Avg. loss: 0.022404, Objective: 0.022404
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.65, NNZs: 1537, Bias: 0.000000, T: 2340, Avg. loss: 0.011698, Objective: 0.011698
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.90, NNZs: 1537, Bias: 0.000000, T: 2808, Avg. loss: 0.004935, Objective: 0.004935
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.38, NNZs: 1537, Bias: 0.000000, T: 3276, Avg. loss: 0.002562, Objective: 0.002562
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.25, NNZs: 1537, Bias: 0.000000, T: 3744, Avg. loss: 0.001137, Objective: 0.001137
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.29, NNZs: 1537, Bias: 0.000000, T: 4212, Avg. loss: 0.000640, Objective: 0.000640
Total training time: 0.00 seconds.
-- Epoch 10
```

```
Norm: 3.12, NNZs: 1537, Bias: 0.000000, T: 4680, Avg. loss: 0.000284, Objective: 0.000284
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.58, NNZs: 1537, Bias: 0.000000, T: 5148, Avg. loss: 0.000205, Objective: 0.000205
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.41, NNZs: 1537, Bias: 0.000000, T: 5616, Avg. loss: 0.000109, Objective: 0.000109
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.29, NNZs: 1537, Bias: 0.000000, T: 6084, Avg. loss: 0.000050, Objective: 0.000050
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.79, NNZs: 1550, Bias: 0.000000, T: 468, Avg. loss: 0.772505, Objective: 0.772505
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.66, NNZs: 1552, Bias: 0.000000, T: 936, Avg. loss: 0.151205, Objective: 0.151205
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.82, NNZs: 1552, Bias: 0.000000, T: 1404, Avg. loss: 0.054299, Objective: 0.054299
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.70, NNZs: 1553, Bias: 0.000000, T: 1872, Avg. loss: 0.021516, Objective: 0.021516
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.23, NNZs: 1553, Bias: 0.000000, T: 2340, Avg. loss: 0.010077, Objective: 0.010077
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.01, NNZs: 1553, Bias: 0.000000, T: 2808, Avg. loss: 0.004118, Objective: 0.004118
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.07, NNZs: 1553, Bias: 0.000000, T: 3276, Avg. loss: 0.002203, Objective: 0.002203
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.73, NNZs: 1553, Bias: 0.000000, T: 3744, Avg. loss: 0.001061, Objective: 0.001061
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.19, NNZs: 1553, Bias: 0.000000, T: 4212, Avg. loss: 0.000461, Objective: 0.000461
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.03, NNZs: 1553, Bias: 0.000000, T: 4680, Avg. loss: 0.000283, Objective: 0.000283
Total training time: 0.00 seconds.
```

```
-- Epoch 11
Norm: 1.08, NNZs: 1553, Bias: 0.000000, T: 5148, Avg. loss: 0.000104, Objective: 0.000104
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.55, NNZs: 1553, Bias: 0.000000, T: 5616, Avg. loss: 0.000066, Objective: 0.000066
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.93, NNZs: 1553, Bias: 0.000000, T: 6084, Avg. loss: 0.000034, Objective: 0.000034
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.19, NNZs: 1527, Bias: 0.000000, T: 468, Avg. loss: 0.585822, Objective: 0.585822
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.62, NNZs: 1531, Bias: 0.000000, T: 936, Avg. loss: 0.082068, Objective: 0.082068
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.00, NNZs: 1533, Bias: 0.000000, T: 1404, Avg. loss: 0.023150, Objective: 0.023150
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.56, NNZs: 1533, Bias: 0.000000, T: 1872, Avg. loss: 0.007250, Objective: 0.007250
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.70, NNZs: 1533, Bias: 0.000000, T: 2340, Avg. loss: 0.003751, Objective: 0.003751
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.92, NNZs: 1533, Bias: 0.000000, T: 2808, Avg. loss: 0.001531, Objective: 0.001531
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.35, NNZs: 1533, Bias: 0.000000, T: 3276, Avg. loss: 0.000865, Objective: 0.000865
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.68, NNZs: 1533, Bias: 0.000000, T: 3744, Avg. loss: 0.000363, Objective: 0.000363
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.12, NNZs: 1533, Bias: 0.000000, T: 4212, Avg. loss: 0.000134, Objective: 0.000134
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.16, NNZs: 1533, Bias: 0.000000, T: 4680, Avg. loss: 0.000073, Objective: 0.000073
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.33, NNZs: 1533, Bias: 0.000000, T: 5148, Avg. loss: 0.000025, Objective: 0.000025
```

```
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.24, NNZs: 1533, Bias: 0.000000, T: 468, Avg. loss: 0.761739, Objective: 0.761739
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.35, NNZs: 1541, Bias: 0.000000, T: 936, Avg. loss: 0.184442, Objective: 0.184442
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.63, NNZs: 1541, Bias: 0.000000, T: 1404, Avg. loss: 0.075414, Objective: 0.075414
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.28, NNZs: 1541, Bias: 0.000000, T: 1872, Avg. loss: 0.031621, Objective: 0.031621
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.48, NNZs: 1541, Bias: 0.000000, T: 2340, Avg. loss: 0.015096, Objective: 0.015096
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.03, NNZs: 1541, Bias: 0.000000, T: 2808, Avg. loss: 0.007537, Objective: 0.007537
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.58, NNZs: 1541, Bias: 0.000000, T: 3276, Avg. loss: 0.004431, Objective: 0.004431
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.10, NNZs: 1541, Bias: 0.000000, T: 3744, Avg. loss: 0.002172, Objective: 0.002172
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.42, NNZs: 1541, Bias: 0.000000, T: 4212, Avg. loss: 0.001077, Objective: 0.001077
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.51, NNZs: 1541, Bias: 0.000000, T: 4680, Avg. loss: 0.000822, Objective: 0.000822
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.52, NNZs: 1541, Bias: 0.000000, T: 5148, Avg. loss: 0.000437, Objective: 0.000437
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.09, NNZs: 1541, Bias: 0.000000, T: 5616, Avg. loss: 0.000297, Objective: 0.000297
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.17, NNZs: 1541, Bias: 0.000000, T: 6084, Avg. loss: 0.000142, Objective: 0.000142
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.27, NNZs: 1541, Bias: 0.000000, T: 6552, Avg. loss: 0.000117, Objective: 0.000117
```

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00117
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.58, NNZs: 1551, Bias: 0.000000, T: 468, Avg. loss: 0.791303, Objective: 0.791303
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.35, NNZs: 1553, Bias: 0.000000, T: 936, Avg. loss: 0.169205, Objective: 0.169205
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.03, NNZs: 1554, Bias: 0.000000, T: 1404, Avg. loss: 0.062931, Objective: 0.062931
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.64, NNZs: 1554, Bias: 0.000000, T: 1872, Avg. loss: 0.027041, Objective: 0.027041
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.65, NNZs: 1554, Bias: 0.000000, T: 2340, Avg. loss: 0.012046, Objective: 0.012046
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.70, NNZs: 1554, Bias: 0.000000, T: 2808, Avg. loss: 0.005292, Objective: 0.005292
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.85, NNZs: 1554, Bias: 0.000000, T: 3276, Avg. loss: 0.003065, Objective: 0.003065
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.63, NNZs: 1554, Bias: 0.000000, T: 3744, Avg. loss: 0.001644, Objective: 0.001644
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.85, NNZs: 1554, Bias: 0.000000, T: 4212, Avg. loss: 0.000877, Objective: 0.000877
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.77, NNZs: 1554, Bias: 0.000000, T: 4680, Avg. loss: 0.000651, Objective: 0.000651
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.44, NNZs: 1554, Bias: 0.000000, T: 5148, Avg. loss: 0.000365, Objective: 0.000365
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.76, NNZs: 1554, Bias: 0.000000, T: 5616, Avg. loss: 0.000155, Objective: 0.000155
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.55, NNZs: 1554, Bias: 0.000000, T: 6084, Avg. loss: 0.000083, Objective: 0.000083
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
```

```
-- Epoch 1
Norm: 1.32, NNZs: 1472, Bias: 0.000000, T: 468, Avg. loss: 0.552183, Objective: 0.55
2183
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.32, NNZs: 1475, Bias: 0.000000, T: 936, Avg. loss: 0.063686, Objective: 0.06
3686
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.47, NNZs: 1478, Bias: 0.000000, T: 1404, Avg. loss: 0.016800, Objective: 0.0
16800
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.15, NNZs: 1478, Bias: 0.000000, T: 1872, Avg. loss: 0.004095, Objective: 0.0
04095
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.49, NNZs: 1478, Bias: 0.000000, T: 2340, Avg. loss: 0.002318, Objective: 0.0
02318
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.58, NNZs: 1478, Bias: 0.000000, T: 2808, Avg. loss: 0.000586, Objective: 0.0
00586
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.29, NNZs: 1478, Bias: 0.000000, T: 3276, Avg. loss: 0.000420, Objective: 0.0
00420
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.19, NNZs: 1478, Bias: 0.000000, T: 3744, Avg. loss: 0.000192, Objective: 0.0
00192
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.47, NNZs: 1478, Bias: 0.000000, T: 4212, Avg. loss: 0.000058, Objective: 0.0
00058
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.15, NNZs: 1478, Bias: 0.000000, T: 4680, Avg. loss: 0.000036, Objective: 0.0
00036
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.11, NNZs: 1478, Bias: 0.000000, T: 5148, Avg. loss: 0.000010, Objective: 0.0
00010
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.64, NNZs: 1506, Bias: 0.000000, T: 468, Avg. loss: 0.741331, Objective: 0.74
1331
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.48, NNZs: 1508, Bias: 0.000000, T: 936, Avg. loss: 0.172810, Objective: 0.17
2810
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.80, NNZs: 1510, Bias: 0.000000, T: 1404, Avg. loss: 0.057622, Objective: 0.0
57622
```

```
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.82, NNZs: 1510, Bias: 0.000000, T: 1872, Avg. loss: 0.027683, Objective: 0.027683
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.45, NNZs: 1510, Bias: 0.000000, T: 2340, Avg. loss: 0.013067, Objective: 0.013067
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.50, NNZs: 1510, Bias: 0.000000, T: 2808, Avg. loss: 0.006432, Objective: 0.006432
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.53, NNZs: 1510, Bias: 0.000000, T: 3276, Avg. loss: 0.003608, Objective: 0.003608
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.63, NNZs: 1510, Bias: 0.000000, T: 3744, Avg. loss: 0.002084, Objective: 0.002084
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.88, NNZs: 1510, Bias: 0.000000, T: 4212, Avg. loss: 0.000841, Objective: 0.000841
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.62, NNZs: 1510, Bias: 0.000000, T: 4680, Avg. loss: 0.000747, Objective: 0.000747
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.96, NNZs: 1510, Bias: 0.000000, T: 5148, Avg. loss: 0.000388, Objective: 0.000388
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.59, NNZs: 1510, Bias: 0.000000, T: 5616, Avg. loss: 0.000325, Objective: 0.000325
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.69, NNZs: 1510, Bias: 0.000000, T: 6084, Avg. loss: 0.000192, Objective: 0.000192
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.07, NNZs: 1510, Bias: 0.000000, T: 6552, Avg. loss: 0.000147, Objective: 0.000147
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.36, NNZs: 1511, Bias: 0.000000, T: 468, Avg. loss: 0.760446, Objective: 0.760446
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.70, NNZs: 1511, Bias: 0.000000, T: 936, Avg. loss: 0.148532, Objective: 0.148532
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.98, NNZs: 1514, Bias: 0.000000, T: 1404, Avg. loss: 0.050794, Objective: 0.050794
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50794
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.77, NNZs: 1514, Bias: 0.000000, T: 1872, Avg. loss: 0.022234, Objective: 0.022234
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.91, NNZs: 1514, Bias: 0.000000, T: 2340, Avg. loss: 0.009361, Objective: 0.009361
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.62, NNZs: 1514, Bias: 0.000000, T: 2808, Avg. loss: 0.005146, Objective: 0.005146
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.17, NNZs: 1514, Bias: 0.000000, T: 3276, Avg. loss: 0.002339, Objective: 0.002339
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.70, NNZs: 1514, Bias: 0.000000, T: 3744, Avg. loss: 0.001340, Objective: 0.001340
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.94, NNZs: 1514, Bias: 0.000000, T: 4212, Avg. loss: 0.000748, Objective: 0.000748
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.17, NNZs: 1514, Bias: 0.000000, T: 4680, Avg. loss: 0.000617, Objective: 0.000617
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.20, NNZs: 1514, Bias: 0.000000, T: 5148, Avg. loss: 0.000459, Objective: 0.000459
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.89, NNZs: 1514, Bias: 0.000000, T: 5616, Avg. loss: 0.000219, Objective: 0.000219
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.72, NNZs: 1475, Bias: 0.000000, T: 468, Avg. loss: 0.570620, Objective: 0.570620
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.29, NNZs: 1481, Bias: 0.000000, T: 936, Avg. loss: 0.078987, Objective: 0.078987
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.57, NNZs: 1482, Bias: 0.000000, T: 1404, Avg. loss: 0.023958, Objective: 0.023958
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.45, NNZs: 1482, Bias: 0.000000, T: 1872, Avg. loss: 0.007818, Objective: 0.007818
Total training time: 0.00 seconds.
-- Epoch 5
```

```
Norm: 1.57, NNZs: 1482, Bias: 0.000000, T: 2340, Avg. loss: 0.003481, Objective: 0.003481
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.74, NNZs: 1482, Bias: 0.000000, T: 2808, Avg. loss: 0.001195, Objective: 0.001195
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.58, NNZs: 1482, Bias: 0.000000, T: 3276, Avg. loss: 0.000622, Objective: 0.000622
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.08, NNZs: 1482, Bias: 0.000000, T: 3744, Avg. loss: 0.000294, Objective: 0.000294
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.33, NNZs: 1482, Bias: 0.000000, T: 4212, Avg. loss: 0.000113, Objective: 0.000113
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.04, NNZs: 1482, Bias: 0.000000, T: 4680, Avg. loss: 0.000066, Objective: 0.000066
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.44, NNZs: 1482, Bias: 0.000000, T: 5148, Avg. loss: 0.000020, Objective: 0.000020
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.30, NNZs: 1496, Bias: 0.000000, T: 468, Avg. loss: 0.746026, Objective: 0.746026
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.73, NNZs: 1496, Bias: 0.000000, T: 936, Avg. loss: 0.171599, Objective: 0.171599
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.11, NNZs: 1498, Bias: 0.000000, T: 1404, Avg. loss: 0.068006, Objective: 0.068006
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.55, NNZs: 1498, Bias: 0.000000, T: 1872, Avg. loss: 0.031833, Objective: 0.031833
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.40, NNZs: 1498, Bias: 0.000000, T: 2340, Avg. loss: 0.011833, Objective: 0.011833
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.89, NNZs: 1498, Bias: 0.000000, T: 2808, Avg. loss: 0.006963, Objective: 0.006963
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.70, NNZs: 1498, Bias: 0.000000, T: 3276, Avg. loss: 0.003606, Objective: 0.003606
Total training time: 0.00 seconds.
```

```
-- Epoch 8
Norm: 3.23, NNZs: 1498, Bias: 0.000000, T: 3744, Avg. loss: 0.002387, Objective: 0.002387
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.26, NNZs: 1498, Bias: 0.000000, T: 4212, Avg. loss: 0.001090, Objective: 0.001090
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.37, NNZs: 1498, Bias: 0.000000, T: 4680, Avg. loss: 0.000979, Objective: 0.000979
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.13, NNZs: 1498, Bias: 0.000000, T: 5148, Avg. loss: 0.000429, Objective: 0.000429
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.36, NNZs: 1498, Bias: 0.000000, T: 5616, Avg. loss: 0.000378, Objective: 0.000378
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.36, NNZs: 1498, Bias: 0.000000, T: 6084, Avg. loss: 0.000210, Objective: 0.000210
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.25, NNZs: 1498, Bias: 0.000000, T: 6552, Avg. loss: 0.000183, Objective: 0.000183
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.75, NNZs: 1519, Bias: 0.000000, T: 468, Avg. loss: 0.770072, Objective: 0.770072
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.14, NNZs: 1519, Bias: 0.000000, T: 936, Avg. loss: 0.153449, Objective: 0.153449
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.72, NNZs: 1519, Bias: 0.000000, T: 1404, Avg. loss: 0.055433, Objective: 0.055433
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.37, NNZs: 1519, Bias: 0.000000, T: 1872, Avg. loss: 0.021970, Objective: 0.021970
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.73, NNZs: 1519, Bias: 0.000000, T: 2340, Avg. loss: 0.011175, Objective: 0.011175
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.42, NNZs: 1519, Bias: 0.000000, T: 2808, Avg. loss: 0.005127, Objective: 0.005127
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.36, NNZs: 1519, Bias: 0.000000, T: 3276, Avg. loss: 0.002768, Objective: 0.002768
```

```
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.59, NNZs: 1519, Bias: 0.000000, T: 3744, Avg. loss: 0.001614, Objective: 0.001614
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.70, NNZs: 1519, Bias: 0.000000, T: 4212, Avg. loss: 0.000761, Objective: 0.000761
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.68, NNZs: 1519, Bias: 0.000000, T: 4680, Avg. loss: 0.000631, Objective: 0.000631
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.12, NNZs: 1519, Bias: 0.000000, T: 5148, Avg. loss: 0.000359, Objective: 0.000359
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.29, NNZs: 1519, Bias: 0.000000, T: 5616, Avg. loss: 0.000162, Objective: 0.000162
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.22, NNZs: 1519, Bias: 0.000000, T: 6084, Avg. loss: 0.000100, Objective: 0.000100
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.66, NNZs: 1536, Bias: 0.000000, T: 468, Avg. loss: 0.578926, Objective: 0.578926
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.36, NNZs: 1541, Bias: 0.000000, T: 936, Avg. loss: 0.074026, Objective: 0.074026
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.03, NNZs: 1543, Bias: 0.000000, T: 1404, Avg. loss: 0.018370, Objective: 0.018370
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.25, NNZs: 1547, Bias: 0.000000, T: 1872, Avg. loss: 0.006368, Objective: 0.006368
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.89, NNZs: 1547, Bias: 0.000000, T: 2340, Avg. loss: 0.002067, Objective: 0.002067
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.66, NNZs: 1547, Bias: 0.000000, T: 2808, Avg. loss: 0.000750, Objective: 0.000750
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.36, NNZs: 1547, Bias: 0.000000, T: 3276, Avg. loss: 0.000328, Objective: 0.000328
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.29, NNZs: 1547, Bias: 0.000000, T: 3744, Avg. loss: 0.000175, Objective: 0.000175
```

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00175
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.54, NNZs: 1547, Bias: 0.000000, T: 4212, Avg. loss: 0.000053, Objective: 0.000053
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.22, NNZs: 1547, Bias: 0.000000, T: 4680, Avg. loss: 0.000027, Objective: 0.000027
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.26, NNZs: 1547, Bias: 0.000000, T: 5148, Avg. loss: 0.000009, Objective: 0.000009
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.44, NNZs: 1581, Bias: 0.000000, T: 468, Avg. loss: 0.738842, Objective: 0.738842
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.11, NNZs: 1586, Bias: 0.000000, T: 936, Avg. loss: 0.171131, Objective: 0.171131
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.56, NNZs: 1586, Bias: 0.000000, T: 1404, Avg. loss: 0.066483, Objective: 0.066483
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.97, NNZs: 1586, Bias: 0.000000, T: 1872, Avg. loss: 0.030961, Objective: 0.030961
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.28, NNZs: 1586, Bias: 0.000000, T: 2340, Avg. loss: 0.015704, Objective: 0.015704
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.85, NNZs: 1586, Bias: 0.000000, T: 2808, Avg. loss: 0.010254, Objective: 0.010254
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.69, NNZs: 1586, Bias: 0.000000, T: 3276, Avg. loss: 0.005178, Objective: 0.005178
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.11, NNZs: 1586, Bias: 0.000000, T: 3744, Avg. loss: 0.003250, Objective: 0.003250
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.75, NNZs: 1586, Bias: 0.000000, T: 4212, Avg. loss: 0.001426, Objective: 0.001426
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.21, NNZs: 1586, Bias: 0.000000, T: 4680, Avg. loss: 0.001263, Objective: 0.001263
Total training time: 0.00 seconds.
-- Epoch 11
```

```
Norm: 2.94, NNZs: 1586, Bias: 0.000000, T: 5148, Avg. loss: 0.000654, Objective: 0.000654
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.28, NNZs: 1586, Bias: 0.000000, T: 5616, Avg. loss: 0.000554, Objective: 0.000554
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.82, NNZs: 1586, Bias: 0.000000, T: 6084, Avg. loss: 0.000294, Objective: 0.000294
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.66, NNZs: 1586, Bias: 0.000000, T: 6552, Avg. loss: 0.000244, Objective: 0.000244
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.96, NNZs: 1579, Bias: 0.000000, T: 468, Avg. loss: 0.775263, Objective: 0.775263
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.72, NNZs: 1584, Bias: 0.000000, T: 936, Avg. loss: 0.162024, Objective: 0.162024
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.34, NNZs: 1584, Bias: 0.000000, T: 1404, Avg. loss: 0.059058, Objective: 0.059058
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.22, NNZs: 1586, Bias: 0.000000, T: 1872, Avg. loss: 0.029004, Objective: 0.029004
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.72, NNZs: 1586, Bias: 0.000000, T: 2340, Avg. loss: 0.014697, Objective: 0.014697
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.21, NNZs: 1586, Bias: 0.000000, T: 2808, Avg. loss: 0.007038, Objective: 0.007038
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.94, NNZs: 1586, Bias: 0.000000, T: 3276, Avg. loss: 0.003723, Objective: 0.003723
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.05, NNZs: 1586, Bias: 0.000000, T: 3744, Avg. loss: 0.002526, Objective: 0.002526
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.48, NNZs: 1586, Bias: 0.000000, T: 4212, Avg. loss: 0.001530, Objective: 0.001530
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.98, NNZs: 1586, Bias: 0.000000, T: 4680, Avg. loss: 0.001305, Objective: 0.001305
Total training time: 0.00 seconds.
```

```
-- Epoch 11
Norm: 1.59, NNZs: 1586, Bias: 0.000000, T: 5148, Avg. loss: 0.000800, Objective: 0.000800
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.69, NNZs: 1586, Bias: 0.000000, T: 5616, Avg. loss: 0.000476, Objective: 0.000476
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.79, NNZs: 1586, Bias: 0.000000, T: 6084, Avg. loss: 0.000279, Objective: 0.000279
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.74, NNZs: 1469, Bias: -0.390319, T: 468, Avg. loss: 0.498233, Objective: 0.498233
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.14, NNZs: 1483, Bias: -0.352526, T: 936, Avg. loss: 0.082125, Objective: 0.082125
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.47, NNZs: 1486, Bias: -0.369890, T: 1404, Avg. loss: 0.019073, Objective: 0.019073
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.19, NNZs: 1486, Bias: -0.367177, T: 1872, Avg. loss: 0.004357, Objective: 0.004357
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.20, NNZs: 1486, Bias: -0.367918, T: 2340, Avg. loss: 0.001460, Objective: 0.001460
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.87, NNZs: 1486, Bias: -0.368454, T: 2808, Avg. loss: 0.000467, Objective: 0.000467
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.28, NNZs: 1486, Bias: -0.368417, T: 3276, Avg. loss: 0.000163, Objective: 0.000163
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.94, NNZs: 1486, Bias: -0.368371, T: 3744, Avg. loss: 0.000047, Objective: 0.000047
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.48, NNZs: 1486, Bias: -0.368362, T: 4212, Avg. loss: 0.000011, Objective: 0.000011
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 3.11, NNZs: 1486, Bias: -0.368372, T: 4680, Avg. loss: 0.000004, Objective: 0.000004
Total training time: 0.00 seconds.
Convergence after 10 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.32, NNZs: 1526, Bias: -0.265522, T: 468, Avg. loss: 0.714142, Objective: 0.714142
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14142
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.41, NNZs: 1531, Bias: -0.191339, T: 936, Avg. loss: 0.196124, Objective: 0.196124
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.21, NNZs: 1531, Bias: -0.227724, T: 1404, Avg. loss: 0.063998, Objective: 0.063998
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.64, NNZs: 1531, Bias: -0.225231, T: 1872, Avg. loss: 0.025743, Objective: 0.025743
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.55, NNZs: 1531, Bias: -0.231990, T: 2340, Avg. loss: 0.012568, Objective: 0.012568
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.10, NNZs: 1531, Bias: -0.232227, T: 2808, Avg. loss: 0.005214, Objective: 0.005214
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.26, NNZs: 1531, Bias: -0.232811, T: 3276, Avg. loss: 0.002678, Objective: 0.002678
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.37, NNZs: 1531, Bias: -0.232675, T: 3744, Avg. loss: 0.001144, Objective: 0.001144
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.43, NNZs: 1531, Bias: -0.233001, T: 4212, Avg. loss: 0.000628, Objective: 0.000628
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 3.15, NNZs: 1531, Bias: -0.232633, T: 4680, Avg. loss: 0.000271, Objective: 0.000271
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.77, NNZs: 1531, Bias: -0.232985, T: 5148, Avg. loss: 0.000205, Objective: 0.000205
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.24, NNZs: 1531, Bias: -0.233068, T: 5616, Avg. loss: 0.000100, Objective: 0.000100
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.43, NNZs: 1531, Bias: -0.233001, T: 6084, Avg. loss: 0.000045, Objective: 0.000045
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.68, NNZs: 1550, Bias: -0.153067, T: 468, Avg. loss: 0.772635, Objective: 0.772635
Total training time: 0.00 seconds.
-- Epoch 2
```

Norm: 1.67, NNZs: 1550, Bias: -0.070661, T: 936, Avg. loss: 0.149661, Objective: 0.149661
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.86, NNZs: 1551, Bias: -0.107904, T: 1404, Avg. loss: 0.054768, Objective: 0.054768
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.55, NNZs: 1551, Bias: -0.101373, T: 1872, Avg. loss: 0.021681, Objective: 0.021681
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.22, NNZs: 1552, Bias: -0.105904, T: 2340, Avg. loss: 0.009829, Objective: 0.009829
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.00, NNZs: 1552, Bias: -0.104972, T: 2808, Avg. loss: 0.004051, Objective: 0.004051
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.11, NNZs: 1552, Bias: -0.103912, T: 3276, Avg. loss: 0.002133, Objective: 0.002133
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.64, NNZs: 1552, Bias: -0.104010, T: 3744, Avg. loss: 0.001006, Objective: 0.001006
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.12, NNZs: 1552, Bias: -0.104526, T: 4212, Avg. loss: 0.000439, Objective: 0.000439
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 0.94, NNZs: 1552, Bias: -0.104218, T: 4680, Avg. loss: 0.000258, Objective: 0.000258
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 0.99, NNZs: 1552, Bias: -0.104279, T: 5148, Avg. loss: 0.000098, Objective: 0.000098
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.60, NNZs: 1552, Bias: -0.104295, T: 5616, Avg. loss: 0.000057, Objective: 0.000057
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.95, NNZs: 1552, Bias: -0.104277, T: 6084, Avg. loss: 0.000030, Objective: 0.000030
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.69, NNZs: 1492, Bias: -0.423668, T: 468, Avg. loss: 0.517627, Objective: 0.517627
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.89, NNZs: 1508, Bias: -0.353326, T: 936, Avg. loss: 0.100652, Objective: 0.100652
Total training time: 0.00 seconds.

```
-- Epoch 3
Norm: 1.63, NNZs: 1510, Bias: -0.392517, T: 1404, Avg. loss: 0.025578, Objective:
0.025578
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.79, NNZs: 1510, Bias: -0.374984, T: 1872, Avg. loss: 0.007982, Objective:
0.007982
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.66, NNZs: 1510, Bias: -0.386149, T: 2340, Avg. loss: 0.003947, Objective:
0.003947
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.76, NNZs: 1510, Bias: -0.382711, T: 2808, Avg. loss: 0.001306, Objective:
0.001306
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.12, NNZs: 1510, Bias: -0.385276, T: 3276, Avg. loss: 0.000789, Objective:
0.000789
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 3.02, NNZs: 1510, Bias: -0.384774, T: 3744, Avg. loss: 0.000288, Objective:
0.000288
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.13, NNZs: 1510, Bias: -0.384892, T: 4212, Avg. loss: 0.000110, Objective:
0.000110
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.48, NNZs: 1510, Bias: -0.384954, T: 4680, Avg. loss: 0.000052, Objective:
0.000052
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.18, NNZs: 1510, Bias: -0.384924, T: 5148, Avg. loss: 0.000017, Objective:
0.000017
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.54, NNZs: 1516, Bias: -0.277238, T: 468, Avg. loss: 0.728120, Objective: 0.7
28120
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.53, NNZs: 1525, Bias: -0.204313, T: 936, Avg. loss: 0.211858, Objective: 0.2
11858
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.39, NNZs: 1525, Bias: -0.233288, T: 1404, Avg. loss: 0.079336, Objective:
0.079336
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.51, NNZs: 1525, Bias: -0.233054, T: 1872, Avg. loss: 0.035263, Objective:
0.035263
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.36, NNZs: 1525, Bias: -0.231660, T: 2340, Avg. loss: 0.016305, Objective:
0.016305
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Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.09, NNZs: 1525, Bias: -0.230708, T: 2808, Avg. loss: 0.008079, Objective:
0.008079
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.41, NNZs: 1525, Bias: -0.232996, T: 3276, Avg. loss: 0.004759, Objective:
0.004759
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.95, NNZs: 1525, Bias: -0.233501, T: 3744, Avg. loss: 0.002360, Objective:
0.002360
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.62, NNZs: 1525, Bias: -0.231526, T: 4212, Avg. loss: 0.001128, Objective:
0.001128
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.53, NNZs: 1525, Bias: -0.231932, T: 4680, Avg. loss: 0.000874, Objective:
0.000874
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.07, NNZs: 1525, Bias: -0.232821, T: 5148, Avg. loss: 0.000434, Objective:
0.000434
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.71, NNZs: 1525, Bias: -0.232663, T: 5616, Avg. loss: 0.000300, Objective:
0.000300
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.95, NNZs: 1525, Bias: -0.232418, T: 6084, Avg. loss: 0.000139, Objective:
0.000139
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.44, NNZs: 1525, Bias: -0.232494, T: 6552, Avg. loss: 0.000117, Objective:
0.000117
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.61, NNZs: 1551, Bias: -0.095375, T: 468, Avg. loss: 0.790210, Objective: 0.7
90210
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.30, NNZs: 1553, Bias: -0.070739, T: 936, Avg. loss: 0.169378, Objective: 0.1
69378
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 0.96, NNZs: 1554, Bias: -0.092976, T: 1404, Avg. loss: 0.062335, Objective:
0.062335
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.91, NNZs: 1554, Bias: -0.086082, T: 1872, Avg. loss: 0.027100, Objective:
0.027100
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.57, NNZs: 1554, Bias: -0.094163, T: 2340, Avg. loss: 0.011819, Objective:
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0.011819
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.13, NNZs: 1554, Bias: -0.093509, T: 2808, Avg. loss: 0.005009, Objective:
0.005009
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.90, NNZs: 1554, Bias: -0.092822, T: 3276, Avg. loss: 0.002901, Objective:
0.002901
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.67, NNZs: 1554, Bias: -0.093347, T: 3744, Avg. loss: 0.001553, Objective:
0.001553
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 3.81, NNZs: 1554, Bias: -0.094337, T: 4212, Avg. loss: 0.000787, Objective:
0.000787
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.86, NNZs: 1554, Bias: -0.094015, T: 4680, Avg. loss: 0.000576, Objective:
0.000576
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.44, NNZs: 1554, Bias: -0.094149, T: 5148, Avg. loss: 0.000324, Objective:
0.000324
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.00, NNZs: 1554, Bias: -0.093858, T: 5616, Avg. loss: 0.000129, Objective:
0.000129
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.53, NNZs: 1554, Bias: -0.094079, T: 6084, Avg. loss: 0.000078, Objective:
0.000078
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.83, NNZs: 1457, Bias: -0.390794, T: 468, Avg. loss: 0.491044, Objective: 0.4
91044
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.42, NNZs: 1470, Bias: -0.355903, T: 936, Avg. loss: 0.077510, Objective: 0.0
77510
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.72, NNZs: 1473, Bias: -0.372918, T: 1404, Avg. loss: 0.017517, Objective:
0.017517
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 0.88, NNZs: 1473, Bias: -0.366141, T: 1872, Avg. loss: 0.004287, Objective:
0.004287
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.73, NNZs: 1473, Bias: -0.372108, T: 2340, Avg. loss: 0.002306, Objective:
0.002306
Total training time: 0.00 seconds.
-- Epoch 6
```

```
Norm: 1.70, NNZs: 1473, Bias: -0.370345, T: 2808, Avg. loss: 0.000575, Objective:
0.000575
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 0.88, NNZs: 1473, Bias: -0.371744, T: 3276, Avg. loss: 0.000404, Objective:
0.000404
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.74, NNZs: 1473, Bias: -0.371541, T: 3744, Avg. loss: 0.000157, Objective:
0.000157
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.72, NNZs: 1473, Bias: -0.371568, T: 4212, Avg. loss: 0.000050, Objective:
0.000050
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.46, NNZs: 1473, Bias: -0.371620, T: 4680, Avg. loss: 0.000028, Objective:
0.000028
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.72, NNZs: 1473, Bias: -0.371589, T: 5148, Avg. loss: 0.000008, Objective:
0.000008
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.52, NNZs: 1494, Bias: -0.247380, T: 468, Avg. loss: 0.713256, Objective: 0.7
13256
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.66, NNZs: 1501, Bias: -0.197606, T: 936, Avg. loss: 0.199805, Objective: 0.1
99805
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.60, NNZs: 1502, Bias: -0.183931, T: 1404, Avg. loss: 0.064491, Objective:
0.064491
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.71, NNZs: 1502, Bias: -0.184880, T: 1872, Avg. loss: 0.030633, Objective:
0.030633
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.32, NNZs: 1502, Bias: -0.185265, T: 2340, Avg. loss: 0.014086, Objective:
0.014086
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.52, NNZs: 1502, Bias: -0.184593, T: 2808, Avg. loss: 0.006857, Objective:
0.006857
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.71, NNZs: 1502, Bias: -0.185327, T: 3276, Avg. loss: 0.003838, Objective:
0.003838
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.65, NNZs: 1502, Bias: -0.185713, T: 3744, Avg. loss: 0.002233, Objective:
0.002233
Total training time: 0.00 seconds.
```

```
-- Epoch 9
Norm: 1.80, NNZs: 1502, Bias: -0.183900, T: 4212, Avg. loss: 0.000877, Objective:
0.000877
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.51, NNZs: 1502, Bias: -0.184305, T: 4680, Avg. loss: 0.000743, Objective:
0.000743
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.05, NNZs: 1502, Bias: -0.185402, T: 5148, Avg. loss: 0.000404, Objective:
0.000404
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.21, NNZs: 1502, Bias: -0.185267, T: 5616, Avg. loss: 0.000323, Objective:
0.000323
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.53, NNZs: 1502, Bias: -0.185034, T: 6084, Avg. loss: 0.000200, Objective:
0.000200
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.24, NNZs: 1502, Bias: -0.184930, T: 6552, Avg. loss: 0.000141, Objective:
0.000141
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.30, NNZs: 1511, Bias: -0.103306, T: 468, Avg. loss: 0.760159, Objective: 0.7
60159
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.61, NNZs: 1512, Bias: -0.115677, T: 936, Avg. loss: 0.146473, Objective: 0.1
46473
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.90, NNZs: 1513, Bias: -0.137742, T: 1404, Avg. loss: 0.049552, Objective:
0.049552
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 3.37, NNZs: 1513, Bias: -0.129350, T: 1872, Avg. loss: 0.021528, Objective:
0.021528
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.26, NNZs: 1513, Bias: -0.136528, T: 2340, Avg. loss: 0.009293, Objective:
0.009293
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.64, NNZs: 1513, Bias: -0.137127, T: 2808, Avg. loss: 0.004694, Objective:
0.004694
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.05, NNZs: 1513, Bias: -0.135907, T: 3276, Avg. loss: 0.002136, Objective:
0.002136
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.62, NNZs: 1513, Bias: -0.136101, T: 3744, Avg. loss: 0.001270, Objective:
0.001270
```

```
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.84, NNZs: 1513, Bias: -0.136993, T: 4212, Avg. loss: 0.000711, Objective:
0.000711
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.99, NNZs: 1513, Bias: -0.136385, T: 4680, Avg. loss: 0.000591, Objective:
0.000591
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.16, NNZs: 1513, Bias: -0.136106, T: 5148, Avg. loss: 0.000432, Objective:
0.000432
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 3.06, NNZs: 1513, Bias: -0.135876, T: 5616, Avg. loss: 0.000202, Objective:
0.000202
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.38, NNZs: 1452, Bias: -0.398522, T: 468, Avg. loss: 0.505690, Objective: 0.5
05690
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.67, NNZs: 1463, Bias: -0.387001, T: 936, Avg. loss: 0.090440, Objective: 0.0
90440
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.11, NNZs: 1463, Bias: -0.394865, T: 1404, Avg. loss: 0.024672, Objective:
0.024672
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.31, NNZs: 1463, Bias: -0.390946, T: 1872, Avg. loss: 0.007587, Objective:
0.007587
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.02, NNZs: 1463, Bias: -0.395567, T: 2340, Avg. loss: 0.003331, Objective:
0.003331
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.43, NNZs: 1463, Bias: -0.393132, T: 2808, Avg. loss: 0.001179, Objective:
0.001179
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.14, NNZs: 1463, Bias: -0.395004, T: 3276, Avg. loss: 0.000611, Objective:
0.000611
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 0.78, NNZs: 1463, Bias: -0.394660, T: 3744, Avg. loss: 0.000282, Objective:
0.000282
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.03, NNZs: 1463, Bias: -0.394624, T: 4212, Avg. loss: 0.000102, Objective:
0.000102
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.26, NNZs: 1463, Bias: -0.394759, T: 4680, Avg. loss: 0.000060, Objective:
```

```
0.000060
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.30, NNZs: 1463, Bias: -0.394703, T: 5148, Avg. loss: 0.000019, Objective:
0.000019
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.04, NNZs: 1498, Bias: -0.299330, T: 468, Avg. loss: 0.719488, Objective: 0.7
19488
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.77, NNZs: 1500, Bias: -0.211550, T: 936, Avg. loss: 0.194570, Objective: 0.1
94570
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.87, NNZs: 1501, Bias: -0.192091, T: 1404, Avg. loss: 0.075087, Objective:
0.075087
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.63, NNZs: 1501, Bias: -0.209930, T: 1872, Avg. loss: 0.034134, Objective:
0.034134
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.30, NNZs: 1501, Bias: -0.214819, T: 2340, Avg. loss: 0.012916, Objective:
0.012916
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.92, NNZs: 1501, Bias: -0.213905, T: 2808, Avg. loss: 0.007239, Objective:
0.007239
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.86, NNZs: 1501, Bias: -0.217965, T: 3276, Avg. loss: 0.003682, Objective:
0.003682
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 3.06, NNZs: 1501, Bias: -0.217471, T: 3744, Avg. loss: 0.002418, Objective:
0.002418
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.17, NNZs: 1501, Bias: -0.215447, T: 4212, Avg. loss: 0.001111, Objective:
0.001111
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.33, NNZs: 1501, Bias: -0.216264, T: 4680, Avg. loss: 0.000961, Objective:
0.000961
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 0.91, NNZs: 1501, Bias: -0.217259, T: 5148, Avg. loss: 0.000424, Objective:
0.000424
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.17, NNZs: 1501, Bias: -0.217109, T: 5616, Avg. loss: 0.000382, Objective:
0.000382
Total training time: 0.00 seconds.
-- Epoch 13
```

Norm: 2.29, NNZs: 1501, Bias: -0.216853, T: 6084, Avg. loss: 0.000214, Objective: 0.000214
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.37, NNZs: 1501, Bias: -0.216868, T: 6552, Avg. loss: 0.000174, Objective: 0.000174
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.73, NNZs: 1519, Bias: -0.075806, T: 468, Avg. loss: 0.769476, Objective: 0.769476
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.17, NNZs: 1519, Bias: -0.097819, T: 936, Avg. loss: 0.153515, Objective: 0.153515
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.78, NNZs: 1519, Bias: -0.098017, T: 1404, Avg. loss: 0.055861, Objective: 0.055861
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.44, NNZs: 1519, Bias: -0.088899, T: 1872, Avg. loss: 0.021577, Objective: 0.021577
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.80, NNZs: 1519, Bias: -0.095977, T: 2340, Avg. loss: 0.011223, Objective: 0.011223
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.42, NNZs: 1519, Bias: -0.094288, T: 2808, Avg. loss: 0.005063, Objective: 0.005063
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.42, NNZs: 1519, Bias: -0.095011, T: 3276, Avg. loss: 0.002745, Objective: 0.002745
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.49, NNZs: 1519, Bias: -0.095229, T: 3744, Avg. loss: 0.001571, Objective: 0.001571
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.61, NNZs: 1519, Bias: -0.096238, T: 4212, Avg. loss: 0.000734, Objective: 0.000734
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.75, NNZs: 1519, Bias: -0.095487, T: 4680, Avg. loss: 0.000594, Objective: 0.000594
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.11, NNZs: 1519, Bias: -0.095381, T: 5148, Avg. loss: 0.000344, Objective: 0.000344
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.07, NNZs: 1519, Bias: -0.095201, T: 5616, Avg. loss: 0.000152, Objective: 0.000152
Total training time: 0.00 seconds.

```
-- Epoch 13
Norm: 1.16, NNZs: 1519, Bias: -0.095495, T: 6084, Avg. loss: 0.000095, Objective:
0.000095
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.33, NNZs: 1527, Bias: -0.375618, T: 468, Avg. loss: 0.512577, Objective: 0.5
12577
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.41, NNZs: 1532, Bias: -0.364790, T: 936, Avg. loss: 0.090941, Objective: 0.0
90941
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.54, NNZs: 1534, Bias: -0.376831, T: 1404, Avg. loss: 0.021166, Objective:
0.021166
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.01, NNZs: 1534, Bias: -0.372679, T: 1872, Avg. loss: 0.007208, Objective:
0.007208
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.95, NNZs: 1534, Bias: -0.376080, T: 2340, Avg. loss: 0.002240, Objective:
0.002240
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.49, NNZs: 1534, Bias: -0.374655, T: 2808, Avg. loss: 0.000881, Objective:
0.000881
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.46, NNZs: 1534, Bias: -0.375582, T: 3276, Avg. loss: 0.000357, Objective:
0.000357
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.70, NNZs: 1534, Bias: -0.375419, T: 3744, Avg. loss: 0.000181, Objective:
0.000181
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.22, NNZs: 1534, Bias: -0.375292, T: 4212, Avg. loss: 0.000047, Objective:
0.000047
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.06, NNZs: 1534, Bias: -0.375382, T: 4680, Avg. loss: 0.000028, Objective:
0.000028
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.10, NNZs: 1534, Bias: -0.375359, T: 5148, Avg. loss: 0.000008, Objective:
0.000008
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.37, NNZs: 1575, Bias: -0.295985, T: 468, Avg. loss: 0.710548, Objective: 0.7
10548
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.98, NNZs: 1581, Bias: -0.216396, T: 936, Avg. loss: 0.191515, Objective: 0.1
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91515
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.61, NNZs: 1582, Bias: -0.208297, T: 1404, Avg. loss: 0.071496, Objective:
0.071496
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.99, NNZs: 1582, Bias: -0.216435, T: 1872, Avg. loss: 0.032887, Objective:
0.032887
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.34, NNZs: 1582, Bias: -0.218042, T: 2340, Avg. loss: 0.016354, Objective:
0.016354
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.72, NNZs: 1582, Bias: -0.219272, T: 2808, Avg. loss: 0.010562, Objective:
0.010562
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.73, NNZs: 1582, Bias: -0.222903, T: 3276, Avg. loss: 0.005210, Objective:
0.005210
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.25, NNZs: 1582, Bias: -0.221989, T: 3744, Avg. loss: 0.003224, Objective:
0.003224
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.29, NNZs: 1582, Bias: -0.220082, T: 4212, Avg. loss: 0.001428, Objective:
0.001428
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.35, NNZs: 1582, Bias: -0.220349, T: 4680, Avg. loss: 0.001261, Objective:
0.001261
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.73, NNZs: 1582, Bias: -0.221962, T: 5148, Avg. loss: 0.000645, Objective:
0.000645
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.04, NNZs: 1582, Bias: -0.221617, T: 5616, Avg. loss: 0.000543, Objective:
0.000543
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.98, NNZs: 1582, Bias: -0.221268, T: 6084, Avg. loss: 0.000290, Objective:
0.000290
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.54, NNZs: 1582, Bias: -0.221209, T: 6552, Avg. loss: 0.000236, Objective:
0.000236
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.97, NNZs: 1585, Bias: -0.075123, T: 468, Avg. loss: 0.775356, Objective: 0.7
75356
Total training time: 0.00 seconds.
-- Epoch 2
```

Norm: 1.66, NNZs: 1589, Bias: -0.115264, T: 936, Avg. loss: 0.161401, Objective: 0.161401
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.31, NNZs: 1589, Bias: -0.098966, T: 1404, Avg. loss: 0.058711, Objective: 0.058711
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.24, NNZs: 1591, Bias: -0.090319, T: 1872, Avg. loss: 0.028545, Objective: 0.028545
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.66, NNZs: 1591, Bias: -0.098929, T: 2340, Avg. loss: 0.014426, Objective: 0.014426
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.16, NNZs: 1591, Bias: -0.095986, T: 2808, Avg. loss: 0.006882, Objective: 0.006882
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.95, NNZs: 1591, Bias: -0.095006, T: 3276, Avg. loss: 0.003725, Objective: 0.003725
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.08, NNZs: 1591, Bias: -0.095809, T: 3744, Avg. loss: 0.002431, Objective: 0.002431
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.45, NNZs: 1591, Bias: -0.096335, T: 4212, Avg. loss: 0.001506, Objective: 0.001506
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.55, NNZs: 1591, Bias: -0.096481, T: 4680, Avg. loss: 0.001233, Objective: 0.001233
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.80, NNZs: 1591, Bias: -0.095805, T: 5148, Avg. loss: 0.000752, Objective: 0.000752
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.03, NNZs: 1591, Bias: -0.095798, T: 5616, Avg. loss: 0.000452, Objective: 0.000452
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.56, NNZs: 1591, Bias: -0.096255, T: 6084, Avg. loss: 0.000256, Objective: 0.000256
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.16, NNZs: 1507, Bias: 0.000000, T: 468, Avg. loss: 0.561863, Objective: 0.561863
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.12, NNZs: 1508, Bias: 0.000000, T: 936, Avg. loss: 0.065525, Objective: 0.065525
Total training time: 0.00 seconds.

```
-- Epoch 3
Norm: 1.24, NNZs: 1508, Bias: 0.000000, T: 1404, Avg. loss: 0.016479, Objective: 0.016479
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.41, NNZs: 1508, Bias: 0.000000, T: 1872, Avg. loss: 0.004067, Objective: 0.004067
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.47, NNZs: 1508, Bias: 0.000000, T: 2340, Avg. loss: 0.001209, Objective: 0.001209
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.94, NNZs: 1508, Bias: 0.000000, T: 2808, Avg. loss: 0.000371, Objective: 0.000371
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.47, NNZs: 1508, Bias: 0.000000, T: 3276, Avg. loss: 0.000129, Objective: 0.000129
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.10, NNZs: 1508, Bias: 0.000000, T: 3744, Avg. loss: 0.000040, Objective: 0.000040
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.32, NNZs: 1508, Bias: 0.000000, T: 4212, Avg. loss: 0.000010, Objective: 0.000010
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.12, NNZs: 1508, Bias: 0.000000, T: 4680, Avg. loss: 0.000004, Objective: 0.000004
Total training time: 0.00 seconds.
Convergence after 10 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.05, NNZs: 1535, Bias: 0.000000, T: 468, Avg. loss: 0.748456, Objective: 0.748456
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.20, NNZs: 1535, Bias: 0.000000, T: 936, Avg. loss: 0.166319, Objective: 0.166319
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.43, NNZs: 1537, Bias: 0.000000, T: 1404, Avg. loss: 0.056108, Objective: 0.056108
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.76, NNZs: 1537, Bias: 0.000000, T: 1872, Avg. loss: 0.022404, Objective: 0.022404
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.65, NNZs: 1537, Bias: 0.000000, T: 2340, Avg. loss: 0.011698, Objective: 0.011698
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.90, NNZs: 1537, Bias: 0.000000, T: 2808, Avg. loss: 0.004935, Objective: 0.004935
```

```
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.38, NNZs: 1537, Bias: 0.000000, T: 3276, Avg. loss: 0.002562, Objective: 0.002562
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.25, NNZs: 1537, Bias: 0.000000, T: 3744, Avg. loss: 0.001137, Objective: 0.001137
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.29, NNZs: 1537, Bias: 0.000000, T: 4212, Avg. loss: 0.000640, Objective: 0.000640
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 3.12, NNZs: 1537, Bias: 0.000000, T: 4680, Avg. loss: 0.000284, Objective: 0.000284
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.58, NNZs: 1537, Bias: 0.000000, T: 5148, Avg. loss: 0.000205, Objective: 0.000205
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.41, NNZs: 1537, Bias: 0.000000, T: 5616, Avg. loss: 0.000109, Objective: 0.000109
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.29, NNZs: 1537, Bias: 0.000000, T: 6084, Avg. loss: 0.000050, Objective: 0.000050
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.79, NNZs: 1550, Bias: 0.000000, T: 468, Avg. loss: 0.772505, Objective: 0.772505
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.66, NNZs: 1552, Bias: 0.000000, T: 936, Avg. loss: 0.151205, Objective: 0.151205
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.82, NNZs: 1552, Bias: 0.000000, T: 1404, Avg. loss: 0.054299, Objective: 0.054299
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.70, NNZs: 1553, Bias: 0.000000, T: 1872, Avg. loss: 0.021516, Objective: 0.021516
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.23, NNZs: 1553, Bias: 0.000000, T: 2340, Avg. loss: 0.010077, Objective: 0.010077
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.01, NNZs: 1553, Bias: 0.000000, T: 2808, Avg. loss: 0.004118, Objective: 0.004118
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.07, NNZs: 1553, Bias: 0.000000, T: 3276, Avg. loss: 0.002203, Objective: 0.002203
```

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02203
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.73, NNZs: 1553, Bias: 0.000000, T: 3744, Avg. loss: 0.001061, Objective: 0.001061
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.19, NNZs: 1553, Bias: 0.000000, T: 4212, Avg. loss: 0.000461, Objective: 0.000461
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.03, NNZs: 1553, Bias: 0.000000, T: 4680, Avg. loss: 0.000283, Objective: 0.000283
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.08, NNZs: 1553, Bias: 0.000000, T: 5148, Avg. loss: 0.000104, Objective: 0.000104
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.55, NNZs: 1553, Bias: 0.000000, T: 5616, Avg. loss: 0.000066, Objective: 0.000066
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.93, NNZs: 1553, Bias: 0.000000, T: 6084, Avg. loss: 0.000034, Objective: 0.000034
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.19, NNZs: 1527, Bias: 0.000000, T: 468, Avg. loss: 0.585822, Objective: 0.585822
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.62, NNZs: 1531, Bias: 0.000000, T: 936, Avg. loss: 0.082068, Objective: 0.082068
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.00, NNZs: 1533, Bias: 0.000000, T: 1404, Avg. loss: 0.023150, Objective: 0.023150
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.56, NNZs: 1533, Bias: 0.000000, T: 1872, Avg. loss: 0.007250, Objective: 0.007250
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.70, NNZs: 1533, Bias: 0.000000, T: 2340, Avg. loss: 0.003751, Objective: 0.003751
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.92, NNZs: 1533, Bias: 0.000000, T: 2808, Avg. loss: 0.001531, Objective: 0.001531
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.35, NNZs: 1533, Bias: 0.000000, T: 3276, Avg. loss: 0.000865, Objective: 0.000865
Total training time: 0.00 seconds.
-- Epoch 8
```

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Norm: 2.68, NNZs: 1533, Bias: 0.000000, T: 3744, Avg. loss: 0.000363, Objective: 0.000363
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.12, NNZs: 1533, Bias: 0.000000, T: 4212, Avg. loss: 0.000134, Objective: 0.000134
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.16, NNZs: 1533, Bias: 0.000000, T: 4680, Avg. loss: 0.000073, Objective: 0.000073
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.33, NNZs: 1533, Bias: 0.000000, T: 5148, Avg. loss: 0.000025, Objective: 0.000025
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.24, NNZs: 1533, Bias: 0.000000, T: 468, Avg. loss: 0.761739, Objective: 0.761739
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.35, NNZs: 1541, Bias: 0.000000, T: 936, Avg. loss: 0.184442, Objective: 0.184442
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.63, NNZs: 1541, Bias: 0.000000, T: 1404, Avg. loss: 0.075414, Objective: 0.075414
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.28, NNZs: 1541, Bias: 0.000000, T: 1872, Avg. loss: 0.031621, Objective: 0.031621
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.48, NNZs: 1541, Bias: 0.000000, T: 2340, Avg. loss: 0.015096, Objective: 0.015096
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.03, NNZs: 1541, Bias: 0.000000, T: 2808, Avg. loss: 0.007537, Objective: 0.007537
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.58, NNZs: 1541, Bias: 0.000000, T: 3276, Avg. loss: 0.004431, Objective: 0.004431
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.10, NNZs: 1541, Bias: 0.000000, T: 3744, Avg. loss: 0.002172, Objective: 0.002172
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.42, NNZs: 1541, Bias: 0.000000, T: 4212, Avg. loss: 0.001077, Objective: 0.001077
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.51, NNZs: 1541, Bias: 0.000000, T: 4680, Avg. loss: 0.000822, Objective: 0.000822
Total training time: 0.00 seconds.
```

```
-- Epoch 11
Norm: 1.52, NNZs: 1541, Bias: 0.000000, T: 5148, Avg. loss: 0.000437, Objective: 0.000437
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.09, NNZs: 1541, Bias: 0.000000, T: 5616, Avg. loss: 0.000297, Objective: 0.000297
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.17, NNZs: 1541, Bias: 0.000000, T: 6084, Avg. loss: 0.000142, Objective: 0.000142
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.27, NNZs: 1541, Bias: 0.000000, T: 6552, Avg. loss: 0.000117, Objective: 0.000117
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.58, NNZs: 1551, Bias: 0.000000, T: 468, Avg. loss: 0.791303, Objective: 0.791303
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.35, NNZs: 1553, Bias: 0.000000, T: 936, Avg. loss: 0.169205, Objective: 0.169205
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.03, NNZs: 1554, Bias: 0.000000, T: 1404, Avg. loss: 0.062931, Objective: 0.062931
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.64, NNZs: 1554, Bias: 0.000000, T: 1872, Avg. loss: 0.027041, Objective: 0.027041
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.65, NNZs: 1554, Bias: 0.000000, T: 2340, Avg. loss: 0.012046, Objective: 0.012046
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.70, NNZs: 1554, Bias: 0.000000, T: 2808, Avg. loss: 0.005292, Objective: 0.005292
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.85, NNZs: 1554, Bias: 0.000000, T: 3276, Avg. loss: 0.003065, Objective: 0.003065
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.63, NNZs: 1554, Bias: 0.000000, T: 3744, Avg. loss: 0.001644, Objective: 0.001644
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.85, NNZs: 1554, Bias: 0.000000, T: 4212, Avg. loss: 0.000877, Objective: 0.000877
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.77, NNZs: 1554, Bias: 0.000000, T: 4680, Avg. loss: 0.000651, Objective: 0.000651
```

```
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.44, NNZs: 1554, Bias: 0.000000, T: 5148, Avg. loss: 0.000365, Objective: 0.000365
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.76, NNZs: 1554, Bias: 0.000000, T: 5616, Avg. loss: 0.000155, Objective: 0.000155
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.55, NNZs: 1554, Bias: 0.000000, T: 6084, Avg. loss: 0.000083, Objective: 0.000083
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.32, NNZs: 1472, Bias: 0.000000, T: 468, Avg. loss: 0.552183, Objective: 0.552183
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.32, NNZs: 1475, Bias: 0.000000, T: 936, Avg. loss: 0.063686, Objective: 0.063686
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.47, NNZs: 1478, Bias: 0.000000, T: 1404, Avg. loss: 0.016800, Objective: 0.016800
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.15, NNZs: 1478, Bias: 0.000000, T: 1872, Avg. loss: 0.004095, Objective: 0.004095
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.49, NNZs: 1478, Bias: 0.000000, T: 2340, Avg. loss: 0.002318, Objective: 0.002318
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.58, NNZs: 1478, Bias: 0.000000, T: 2808, Avg. loss: 0.000586, Objective: 0.000586
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.29, NNZs: 1478, Bias: 0.000000, T: 3276, Avg. loss: 0.000420, Objective: 0.000420
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.19, NNZs: 1478, Bias: 0.000000, T: 3744, Avg. loss: 0.000192, Objective: 0.000192
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.47, NNZs: 1478, Bias: 0.000000, T: 4212, Avg. loss: 0.000058, Objective: 0.000058
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.15, NNZs: 1478, Bias: 0.000000, T: 4680, Avg. loss: 0.000036, Objective: 0.000036
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.11, NNZs: 1478, Bias: 0.000000, T: 5148, Avg. loss: 0.000010, Objective: 0.000010
```

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00010
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.64, NNZs: 1506, Bias: 0.000000, T: 468, Avg. loss: 0.741331, Objective: 0.741331
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.48, NNZs: 1508, Bias: 0.000000, T: 936, Avg. loss: 0.172810, Objective: 0.172810
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.80, NNZs: 1510, Bias: 0.000000, T: 1404, Avg. loss: 0.057622, Objective: 0.057622
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.82, NNZs: 1510, Bias: 0.000000, T: 1872, Avg. loss: 0.027683, Objective: 0.027683
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.45, NNZs: 1510, Bias: 0.000000, T: 2340, Avg. loss: 0.013067, Objective: 0.013067
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.50, NNZs: 1510, Bias: 0.000000, T: 2808, Avg. loss: 0.006432, Objective: 0.006432
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.53, NNZs: 1510, Bias: 0.000000, T: 3276, Avg. loss: 0.003608, Objective: 0.003608
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.63, NNZs: 1510, Bias: 0.000000, T: 3744, Avg. loss: 0.002084, Objective: 0.002084
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.88, NNZs: 1510, Bias: 0.000000, T: 4212, Avg. loss: 0.000841, Objective: 0.000841
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.62, NNZs: 1510, Bias: 0.000000, T: 4680, Avg. loss: 0.000747, Objective: 0.000747
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.96, NNZs: 1510, Bias: 0.000000, T: 5148, Avg. loss: 0.000388, Objective: 0.000388
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.59, NNZs: 1510, Bias: 0.000000, T: 5616, Avg. loss: 0.000325, Objective: 0.000325
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.69, NNZs: 1510, Bias: 0.000000, T: 6084, Avg. loss: 0.000192, Objective: 0.000192
Total training time: 0.00 seconds.
-- Epoch 14
```

Norm: 2.07, NNZs: 1510, Bias: 0.000000, T: 6552, Avg. loss: 0.000147, Objective: 0.000147
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.36, NNZs: 1511, Bias: 0.000000, T: 468, Avg. loss: 0.760446, Objective: 0.760446
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.70, NNZs: 1511, Bias: 0.000000, T: 936, Avg. loss: 0.148532, Objective: 0.148532
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.98, NNZs: 1514, Bias: 0.000000, T: 1404, Avg. loss: 0.050794, Objective: 0.050794
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.77, NNZs: 1514, Bias: 0.000000, T: 1872, Avg. loss: 0.022234, Objective: 0.022234
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.91, NNZs: 1514, Bias: 0.000000, T: 2340, Avg. loss: 0.009361, Objective: 0.009361
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.62, NNZs: 1514, Bias: 0.000000, T: 2808, Avg. loss: 0.005146, Objective: 0.005146
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.17, NNZs: 1514, Bias: 0.000000, T: 3276, Avg. loss: 0.002339, Objective: 0.002339
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.70, NNZs: 1514, Bias: 0.000000, T: 3744, Avg. loss: 0.001340, Objective: 0.001340
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.94, NNZs: 1514, Bias: 0.000000, T: 4212, Avg. loss: 0.000748, Objective: 0.000748
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.17, NNZs: 1514, Bias: 0.000000, T: 4680, Avg. loss: 0.000617, Objective: 0.000617
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.20, NNZs: 1514, Bias: 0.000000, T: 5148, Avg. loss: 0.000459, Objective: 0.000459
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.89, NNZs: 1514, Bias: 0.000000, T: 5616, Avg. loss: 0.000219, Objective: 0.000219
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.72, NNZs: 1475, Bias: 0.000000, T: 468, Avg. loss: 0.570620, Objective: 0.570620

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Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.29, NNZs: 1481, Bias: 0.000000, T: 936, Avg. loss: 0.078987, Objective: 0.078987
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.57, NNZs: 1482, Bias: 0.000000, T: 1404, Avg. loss: 0.023958, Objective: 0.023958
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.45, NNZs: 1482, Bias: 0.000000, T: 1872, Avg. loss: 0.007818, Objective: 0.007818
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.57, NNZs: 1482, Bias: 0.000000, T: 2340, Avg. loss: 0.003481, Objective: 0.003481
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.74, NNZs: 1482, Bias: 0.000000, T: 2808, Avg. loss: 0.001195, Objective: 0.001195
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.58, NNZs: 1482, Bias: 0.000000, T: 3276, Avg. loss: 0.000622, Objective: 0.000622
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.08, NNZs: 1482, Bias: 0.000000, T: 3744, Avg. loss: 0.000294, Objective: 0.000294
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.33, NNZs: 1482, Bias: 0.000000, T: 4212, Avg. loss: 0.000113, Objective: 0.000113
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.04, NNZs: 1482, Bias: 0.000000, T: 4680, Avg. loss: 0.000066, Objective: 0.000066
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.44, NNZs: 1482, Bias: 0.000000, T: 5148, Avg. loss: 0.000020, Objective: 0.000020
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.30, NNZs: 1496, Bias: 0.000000, T: 468, Avg. loss: 0.746026, Objective: 0.746026
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.73, NNZs: 1496, Bias: 0.000000, T: 936, Avg. loss: 0.171599, Objective: 0.171599
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.11, NNZs: 1498, Bias: 0.000000, T: 1404, Avg. loss: 0.068006, Objective: 0.068006
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.55, NNZs: 1498, Bias: 0.000000, T: 1872, Avg. loss: 0.031833, Objective: 0.031833
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31833
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.40, NNZs: 1498, Bias: 0.000000, T: 2340, Avg. loss: 0.011833, Objective: 0.011833
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.89, NNZs: 1498, Bias: 0.000000, T: 2808, Avg. loss: 0.006963, Objective: 0.006963
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.70, NNZs: 1498, Bias: 0.000000, T: 3276, Avg. loss: 0.003606, Objective: 0.003606
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 3.23, NNZs: 1498, Bias: 0.000000, T: 3744, Avg. loss: 0.002387, Objective: 0.002387
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.26, NNZs: 1498, Bias: 0.000000, T: 4212, Avg. loss: 0.001090, Objective: 0.001090
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.37, NNZs: 1498, Bias: 0.000000, T: 4680, Avg. loss: 0.000979, Objective: 0.000979
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.13, NNZs: 1498, Bias: 0.000000, T: 5148, Avg. loss: 0.000429, Objective: 0.000429
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.36, NNZs: 1498, Bias: 0.000000, T: 5616, Avg. loss: 0.000378, Objective: 0.000378
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.36, NNZs: 1498, Bias: 0.000000, T: 6084, Avg. loss: 0.000210, Objective: 0.000210
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.25, NNZs: 1498, Bias: 0.000000, T: 6552, Avg. loss: 0.000183, Objective: 0.000183
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.75, NNZs: 1519, Bias: 0.000000, T: 468, Avg. loss: 0.770072, Objective: 0.770072
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.14, NNZs: 1519, Bias: 0.000000, T: 936, Avg. loss: 0.153449, Objective: 0.153449
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.72, NNZs: 1519, Bias: 0.000000, T: 1404, Avg. loss: 0.055433, Objective: 0.055433
Total training time: 0.00 seconds.
-- Epoch 4
```

Norm: 2.37, NNZs: 1519, Bias: 0.000000, T: 1872, Avg. loss: 0.021970, Objective: 0.021970
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.73, NNZs: 1519, Bias: 0.000000, T: 2340, Avg. loss: 0.011175, Objective: 0.011175
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.42, NNZs: 1519, Bias: 0.000000, T: 2808, Avg. loss: 0.005127, Objective: 0.005127
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.36, NNZs: 1519, Bias: 0.000000, T: 3276, Avg. loss: 0.002768, Objective: 0.002768
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.59, NNZs: 1519, Bias: 0.000000, T: 3744, Avg. loss: 0.001614, Objective: 0.001614
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.70, NNZs: 1519, Bias: 0.000000, T: 4212, Avg. loss: 0.000761, Objective: 0.000761
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.68, NNZs: 1519, Bias: 0.000000, T: 4680, Avg. loss: 0.000631, Objective: 0.000631
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.12, NNZs: 1519, Bias: 0.000000, T: 5148, Avg. loss: 0.000359, Objective: 0.000359
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.29, NNZs: 1519, Bias: 0.000000, T: 5616, Avg. loss: 0.000162, Objective: 0.000162
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.22, NNZs: 1519, Bias: 0.000000, T: 6084, Avg. loss: 0.000100, Objective: 0.000100
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.66, NNZs: 1536, Bias: 0.000000, T: 468, Avg. loss: 0.578926, Objective: 0.578926
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.36, NNZs: 1541, Bias: 0.000000, T: 936, Avg. loss: 0.074026, Objective: 0.074026
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.03, NNZs: 1543, Bias: 0.000000, T: 1404, Avg. loss: 0.018370, Objective: 0.018370
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.25, NNZs: 1547, Bias: 0.000000, T: 1872, Avg. loss: 0.006368, Objective: 0.006368
Total training time: 0.00 seconds.

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-- Epoch 5
Norm: 1.89, NNZs: 1547, Bias: 0.000000, T: 2340, Avg. loss: 0.002067, Objective: 0.002067
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.66, NNZs: 1547, Bias: 0.000000, T: 2808, Avg. loss: 0.000750, Objective: 0.000750
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.36, NNZs: 1547, Bias: 0.000000, T: 3276, Avg. loss: 0.000328, Objective: 0.000328
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.29, NNZs: 1547, Bias: 0.000000, T: 3744, Avg. loss: 0.000175, Objective: 0.000175
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.54, NNZs: 1547, Bias: 0.000000, T: 4212, Avg. loss: 0.000053, Objective: 0.000053
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.22, NNZs: 1547, Bias: 0.000000, T: 4680, Avg. loss: 0.000027, Objective: 0.000027
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.26, NNZs: 1547, Bias: 0.000000, T: 5148, Avg. loss: 0.000009, Objective: 0.000009
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.44, NNZs: 1581, Bias: 0.000000, T: 468, Avg. loss: 0.738842, Objective: 0.738842
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.11, NNZs: 1586, Bias: 0.000000, T: 936, Avg. loss: 0.171131, Objective: 0.171131
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.56, NNZs: 1586, Bias: 0.000000, T: 1404, Avg. loss: 0.066483, Objective: 0.066483
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.97, NNZs: 1586, Bias: 0.000000, T: 1872, Avg. loss: 0.030961, Objective: 0.030961
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.28, NNZs: 1586, Bias: 0.000000, T: 2340, Avg. loss: 0.015704, Objective: 0.015704
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.85, NNZs: 1586, Bias: 0.000000, T: 2808, Avg. loss: 0.010254, Objective: 0.010254
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.69, NNZs: 1586, Bias: 0.000000, T: 3276, Avg. loss: 0.005178, Objective: 0.005178
```

```
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.11, NNZs: 1586, Bias: 0.000000, T: 3744, Avg. loss: 0.003250, Objective: 0.003250
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.75, NNZs: 1586, Bias: 0.000000, T: 4212, Avg. loss: 0.001426, Objective: 0.001426
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.21, NNZs: 1586, Bias: 0.000000, T: 4680, Avg. loss: 0.001263, Objective: 0.001263
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.94, NNZs: 1586, Bias: 0.000000, T: 5148, Avg. loss: 0.000654, Objective: 0.000654
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.28, NNZs: 1586, Bias: 0.000000, T: 5616, Avg. loss: 0.000554, Objective: 0.000554
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.82, NNZs: 1586, Bias: 0.000000, T: 6084, Avg. loss: 0.000294, Objective: 0.000294
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.66, NNZs: 1586, Bias: 0.000000, T: 6552, Avg. loss: 0.000244, Objective: 0.000244
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.96, NNZs: 1579, Bias: 0.000000, T: 468, Avg. loss: 0.775263, Objective: 0.775263
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.72, NNZs: 1584, Bias: 0.000000, T: 936, Avg. loss: 0.162024, Objective: 0.162024
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.34, NNZs: 1584, Bias: 0.000000, T: 1404, Avg. loss: 0.059058, Objective: 0.059058
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.22, NNZs: 1586, Bias: 0.000000, T: 1872, Avg. loss: 0.029004, Objective: 0.029004
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.72, NNZs: 1586, Bias: 0.000000, T: 2340, Avg. loss: 0.014697, Objective: 0.014697
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.21, NNZs: 1586, Bias: 0.000000, T: 2808, Avg. loss: 0.007038, Objective: 0.007038
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.94, NNZs: 1586, Bias: 0.000000, T: 3276, Avg. loss: 0.003723, Objective: 0.003723
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03723
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.05, NNZs: 1586, Bias: 0.000000, T: 3744, Avg. loss: 0.002526, Objective: 0.002526
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.48, NNZs: 1586, Bias: 0.000000, T: 4212, Avg. loss: 0.001530, Objective: 0.001530
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.98, NNZs: 1586, Bias: 0.000000, T: 4680, Avg. loss: 0.001305, Objective: 0.001305
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.59, NNZs: 1586, Bias: 0.000000, T: 5148, Avg. loss: 0.000800, Objective: 0.000800
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.69, NNZs: 1586, Bias: 0.000000, T: 5616, Avg. loss: 0.000476, Objective: 0.000476
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.79, NNZs: 1586, Bias: 0.000000, T: 6084, Avg. loss: 0.000279, Objective: 0.000279
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.16, NNZs: 1507, Bias: 0.000000, T: 468, Avg. loss: 0.561863, Objective: 0.561863
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.12, NNZs: 1508, Bias: 0.000000, T: 936, Avg. loss: 0.065525, Objective: 0.065525
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.24, NNZs: 1508, Bias: 0.000000, T: 1404, Avg. loss: 0.016479, Objective: 0.016479
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.41, NNZs: 1508, Bias: 0.000000, T: 1872, Avg. loss: 0.004067, Objective: 0.004067
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.47, NNZs: 1508, Bias: 0.000000, T: 2340, Avg. loss: 0.001209, Objective: 0.001209
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.94, NNZs: 1508, Bias: 0.000000, T: 2808, Avg. loss: 0.000371, Objective: 0.000371
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.47, NNZs: 1508, Bias: 0.000000, T: 3276, Avg. loss: 0.000129, Objective: 0.000129
Total training time: 0.00 seconds.
-- Epoch 8
```

```
Norm: 2.10, NNZs: 1508, Bias: 0.000000, T: 3744, Avg. loss: 0.000040, Objective: 0.000040
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.32, NNZs: 1508, Bias: 0.000000, T: 4212, Avg. loss: 0.000010, Objective: 0.000010
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.12, NNZs: 1508, Bias: 0.000000, T: 4680, Avg. loss: 0.000004, Objective: 0.000004
Total training time: 0.00 seconds.
Convergence after 10 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.05, NNZs: 1535, Bias: 0.000000, T: 468, Avg. loss: 0.748456, Objective: 0.748456
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.20, NNZs: 1535, Bias: 0.000000, T: 936, Avg. loss: 0.166319, Objective: 0.166319
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.43, NNZs: 1537, Bias: 0.000000, T: 1404, Avg. loss: 0.056108, Objective: 0.056108
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.76, NNZs: 1537, Bias: 0.000000, T: 1872, Avg. loss: 0.022404, Objective: 0.022404
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.65, NNZs: 1537, Bias: 0.000000, T: 2340, Avg. loss: 0.011698, Objective: 0.011698
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.90, NNZs: 1537, Bias: 0.000000, T: 2808, Avg. loss: 0.004935, Objective: 0.004935
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 2.38, NNZs: 1537, Bias: 0.000000, T: 3276, Avg. loss: 0.002562, Objective: 0.002562
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.25, NNZs: 1537, Bias: 0.000000, T: 3744, Avg. loss: 0.001137, Objective: 0.001137
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.29, NNZs: 1537, Bias: 0.000000, T: 4212, Avg. loss: 0.000640, Objective: 0.000640
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 3.12, NNZs: 1537, Bias: 0.000000, T: 4680, Avg. loss: 0.000284, Objective: 0.000284
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.58, NNZs: 1537, Bias: 0.000000, T: 5148, Avg. loss: 0.000205, Objective: 0.000205
Total training time: 0.00 seconds.
```

```
-- Epoch 12
Norm: 2.41, NNZs: 1537, Bias: 0.000000, T: 5616, Avg. loss: 0.000109, Objective: 0.000109
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.29, NNZs: 1537, Bias: 0.000000, T: 6084, Avg. loss: 0.000050, Objective: 0.000050
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.79, NNZs: 1550, Bias: 0.000000, T: 468, Avg. loss: 0.772505, Objective: 0.772505
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.66, NNZs: 1552, Bias: 0.000000, T: 936, Avg. loss: 0.151205, Objective: 0.151205
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.82, NNZs: 1552, Bias: 0.000000, T: 1404, Avg. loss: 0.054299, Objective: 0.054299
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.70, NNZs: 1553, Bias: 0.000000, T: 1872, Avg. loss: 0.021516, Objective: 0.021516
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.23, NNZs: 1553, Bias: 0.000000, T: 2340, Avg. loss: 0.010077, Objective: 0.010077
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.01, NNZs: 1553, Bias: 0.000000, T: 2808, Avg. loss: 0.004118, Objective: 0.004118
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.07, NNZs: 1553, Bias: 0.000000, T: 3276, Avg. loss: 0.002203, Objective: 0.002203
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.73, NNZs: 1553, Bias: 0.000000, T: 3744, Avg. loss: 0.001061, Objective: 0.001061
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.19, NNZs: 1553, Bias: 0.000000, T: 4212, Avg. loss: 0.000461, Objective: 0.000461
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.03, NNZs: 1553, Bias: 0.000000, T: 4680, Avg. loss: 0.000283, Objective: 0.000283
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.08, NNZs: 1553, Bias: 0.000000, T: 5148, Avg. loss: 0.000104, Objective: 0.000104
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.55, NNZs: 1553, Bias: 0.000000, T: 5616, Avg. loss: 0.000066, Objective: 0.000066
```

```
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.93, NNZs: 1553, Bias: 0.000000, T: 6084, Avg. loss: 0.000034, Objective: 0.000034
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.19, NNZs: 1527, Bias: 0.000000, T: 468, Avg. loss: 0.585822, Objective: 0.585822
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.62, NNZs: 1531, Bias: 0.000000, T: 936, Avg. loss: 0.082068, Objective: 0.082068
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.00, NNZs: 1533, Bias: 0.000000, T: 1404, Avg. loss: 0.023150, Objective: 0.023150
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.56, NNZs: 1533, Bias: 0.000000, T: 1872, Avg. loss: 0.007250, Objective: 0.007250
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.70, NNZs: 1533, Bias: 0.000000, T: 2340, Avg. loss: 0.003751, Objective: 0.003751
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.92, NNZs: 1533, Bias: 0.000000, T: 2808, Avg. loss: 0.001531, Objective: 0.001531
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.35, NNZs: 1533, Bias: 0.000000, T: 3276, Avg. loss: 0.000865, Objective: 0.000865
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.68, NNZs: 1533, Bias: 0.000000, T: 3744, Avg. loss: 0.000363, Objective: 0.000363
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.12, NNZs: 1533, Bias: 0.000000, T: 4212, Avg. loss: 0.000134, Objective: 0.000134
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.16, NNZs: 1533, Bias: 0.000000, T: 4680, Avg. loss: 0.000073, Objective: 0.000073
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.33, NNZs: 1533, Bias: 0.000000, T: 5148, Avg. loss: 0.000025, Objective: 0.000025
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.24, NNZs: 1533, Bias: 0.000000, T: 468, Avg. loss: 0.761739, Objective: 0.761739
Total training time: 0.00 seconds.
-- Epoch 2
```

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Norm: 1.35, NNZs: 1541, Bias: 0.000000, T: 936, Avg. loss: 0.184442, Objective: 0.184442
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.63, NNZs: 1541, Bias: 0.000000, T: 1404, Avg. loss: 0.075414, Objective: 0.075414
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.28, NNZs: 1541, Bias: 0.000000, T: 1872, Avg. loss: 0.031621, Objective: 0.031621
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.48, NNZs: 1541, Bias: 0.000000, T: 2340, Avg. loss: 0.015096, Objective: 0.015096
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.03, NNZs: 1541, Bias: 0.000000, T: 2808, Avg. loss: 0.007537, Objective: 0.007537
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.58, NNZs: 1541, Bias: 0.000000, T: 3276, Avg. loss: 0.004431, Objective: 0.004431
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.10, NNZs: 1541, Bias: 0.000000, T: 3744, Avg. loss: 0.002172, Objective: 0.002172
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.42, NNZs: 1541, Bias: 0.000000, T: 4212, Avg. loss: 0.001077, Objective: 0.001077
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.51, NNZs: 1541, Bias: 0.000000, T: 4680, Avg. loss: 0.000822, Objective: 0.000822
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.52, NNZs: 1541, Bias: 0.000000, T: 5148, Avg. loss: 0.000437, Objective: 0.000437
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.09, NNZs: 1541, Bias: 0.000000, T: 5616, Avg. loss: 0.000297, Objective: 0.000297
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.17, NNZs: 1541, Bias: 0.000000, T: 6084, Avg. loss: 0.000142, Objective: 0.000142
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.27, NNZs: 1541, Bias: 0.000000, T: 6552, Avg. loss: 0.000117, Objective: 0.000117
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.58, NNZs: 1551, Bias: 0.000000, T: 468, Avg. loss: 0.791303, Objective: 0.791303
Total training time: 0.00 seconds.
```

```
-- Epoch 2
Norm: 1.35, NNZs: 1553, Bias: 0.000000, T: 936, Avg. loss: 0.169205, Objective: 0.169205
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.03, NNZs: 1554, Bias: 0.000000, T: 1404, Avg. loss: 0.062931, Objective: 0.062931
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.64, NNZs: 1554, Bias: 0.000000, T: 1872, Avg. loss: 0.027041, Objective: 0.027041
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.65, NNZs: 1554, Bias: 0.000000, T: 2340, Avg. loss: 0.012046, Objective: 0.012046
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.70, NNZs: 1554, Bias: 0.000000, T: 2808, Avg. loss: 0.005292, Objective: 0.005292
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.85, NNZs: 1554, Bias: 0.000000, T: 3276, Avg. loss: 0.003065, Objective: 0.003065
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.63, NNZs: 1554, Bias: 0.000000, T: 3744, Avg. loss: 0.001644, Objective: 0.001644
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.85, NNZs: 1554, Bias: 0.000000, T: 4212, Avg. loss: 0.000877, Objective: 0.000877
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.77, NNZs: 1554, Bias: 0.000000, T: 4680, Avg. loss: 0.000651, Objective: 0.000651
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.44, NNZs: 1554, Bias: 0.000000, T: 5148, Avg. loss: 0.000365, Objective: 0.000365
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.76, NNZs: 1554, Bias: 0.000000, T: 5616, Avg. loss: 0.000155, Objective: 0.000155
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.55, NNZs: 1554, Bias: 0.000000, T: 6084, Avg. loss: 0.000083, Objective: 0.000083
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.32, NNZs: 1472, Bias: 0.000000, T: 468, Avg. loss: 0.552183, Objective: 0.552183
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.32, NNZs: 1475, Bias: 0.000000, T: 936, Avg. loss: 0.063686, Objective: 0.063686
```

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Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.47, NNZs: 1478, Bias: 0.000000, T: 1404, Avg. loss: 0.016800, Objective: 0.016800
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.15, NNZs: 1478, Bias: 0.000000, T: 1872, Avg. loss: 0.004095, Objective: 0.004095
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.49, NNZs: 1478, Bias: 0.000000, T: 2340, Avg. loss: 0.002318, Objective: 0.002318
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.58, NNZs: 1478, Bias: 0.000000, T: 2808, Avg. loss: 0.000586, Objective: 0.000586
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.29, NNZs: 1478, Bias: 0.000000, T: 3276, Avg. loss: 0.000420, Objective: 0.000420
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.19, NNZs: 1478, Bias: 0.000000, T: 3744, Avg. loss: 0.000192, Objective: 0.000192
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.47, NNZs: 1478, Bias: 0.000000, T: 4212, Avg. loss: 0.000058, Objective: 0.000058
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.15, NNZs: 1478, Bias: 0.000000, T: 4680, Avg. loss: 0.000036, Objective: 0.000036
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.11, NNZs: 1478, Bias: 0.000000, T: 5148, Avg. loss: 0.000010, Objective: 0.000010
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.64, NNZs: 1506, Bias: 0.000000, T: 468, Avg. loss: 0.741331, Objective: 0.741331
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.48, NNZs: 1508, Bias: 0.000000, T: 936, Avg. loss: 0.172810, Objective: 0.172810
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.80, NNZs: 1510, Bias: 0.000000, T: 1404, Avg. loss: 0.057622, Objective: 0.057622
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.82, NNZs: 1510, Bias: 0.000000, T: 1872, Avg. loss: 0.027683, Objective: 0.027683
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.45, NNZs: 1510, Bias: 0.000000, T: 2340, Avg. loss: 0.013067, Objective: 0.013067
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13067
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.50, NNZs: 1510, Bias: 0.000000, T: 2808, Avg. loss: 0.006432, Objective: 0.006432
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.53, NNZs: 1510, Bias: 0.000000, T: 3276, Avg. loss: 0.003608, Objective: 0.003608
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.63, NNZs: 1510, Bias: 0.000000, T: 3744, Avg. loss: 0.002084, Objective: 0.002084
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.88, NNZs: 1510, Bias: 0.000000, T: 4212, Avg. loss: 0.000841, Objective: 0.000841
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.62, NNZs: 1510, Bias: 0.000000, T: 4680, Avg. loss: 0.000747, Objective: 0.000747
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.96, NNZs: 1510, Bias: 0.000000, T: 5148, Avg. loss: 0.000388, Objective: 0.000388
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.59, NNZs: 1510, Bias: 0.000000, T: 5616, Avg. loss: 0.000325, Objective: 0.000325
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.69, NNZs: 1510, Bias: 0.000000, T: 6084, Avg. loss: 0.000192, Objective: 0.000192
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 2.07, NNZs: 1510, Bias: 0.000000, T: 6552, Avg. loss: 0.000147, Objective: 0.000147
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.36, NNZs: 1511, Bias: 0.000000, T: 468, Avg. loss: 0.760446, Objective: 0.760446
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.70, NNZs: 1511, Bias: 0.000000, T: 936, Avg. loss: 0.148532, Objective: 0.148532
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.98, NNZs: 1514, Bias: 0.000000, T: 1404, Avg. loss: 0.050794, Objective: 0.050794
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.77, NNZs: 1514, Bias: 0.000000, T: 1872, Avg. loss: 0.022234, Objective: 0.022234
Total training time: 0.00 seconds.
-- Epoch 5
```

```
Norm: 1.91, NNZs: 1514, Bias: 0.000000, T: 2340, Avg. loss: 0.009361, Objective: 0.009361
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.62, NNZs: 1514, Bias: 0.000000, T: 2808, Avg. loss: 0.005146, Objective: 0.005146
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.17, NNZs: 1514, Bias: 0.000000, T: 3276, Avg. loss: 0.002339, Objective: 0.002339
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.70, NNZs: 1514, Bias: 0.000000, T: 3744, Avg. loss: 0.001340, Objective: 0.001340
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.94, NNZs: 1514, Bias: 0.000000, T: 4212, Avg. loss: 0.000748, Objective: 0.000748
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.17, NNZs: 1514, Bias: 0.000000, T: 4680, Avg. loss: 0.000617, Objective: 0.000617
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.20, NNZs: 1514, Bias: 0.000000, T: 5148, Avg. loss: 0.000459, Objective: 0.000459
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.89, NNZs: 1514, Bias: 0.000000, T: 5616, Avg. loss: 0.000219, Objective: 0.000219
Total training time: 0.00 seconds.
Convergence after 12 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.72, NNZs: 1475, Bias: 0.000000, T: 468, Avg. loss: 0.570620, Objective: 0.570620
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.29, NNZs: 1481, Bias: 0.000000, T: 936, Avg. loss: 0.078987, Objective: 0.078987
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.57, NNZs: 1482, Bias: 0.000000, T: 1404, Avg. loss: 0.023958, Objective: 0.023958
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.45, NNZs: 1482, Bias: 0.000000, T: 1872, Avg. loss: 0.007818, Objective: 0.007818
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.57, NNZs: 1482, Bias: 0.000000, T: 2340, Avg. loss: 0.003481, Objective: 0.003481
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.74, NNZs: 1482, Bias: 0.000000, T: 2808, Avg. loss: 0.001195, Objective: 0.001195
Total training time: 0.00 seconds.
```

```
-- Epoch 7
Norm: 1.58, NNZs: 1482, Bias: 0.000000, T: 3276, Avg. loss: 0.000622, Objective: 0.000622
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.08, NNZs: 1482, Bias: 0.000000, T: 3744, Avg. loss: 0.000294, Objective: 0.000294
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.33, NNZs: 1482, Bias: 0.000000, T: 4212, Avg. loss: 0.000113, Objective: 0.000113
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.04, NNZs: 1482, Bias: 0.000000, T: 4680, Avg. loss: 0.000066, Objective: 0.000066
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.44, NNZs: 1482, Bias: 0.000000, T: 5148, Avg. loss: 0.000020, Objective: 0.000020
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.30, NNZs: 1496, Bias: 0.000000, T: 468, Avg. loss: 0.746026, Objective: 0.746026
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.73, NNZs: 1496, Bias: 0.000000, T: 936, Avg. loss: 0.171599, Objective: 0.171599
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.11, NNZs: 1498, Bias: 0.000000, T: 1404, Avg. loss: 0.068006, Objective: 0.068006
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.55, NNZs: 1498, Bias: 0.000000, T: 1872, Avg. loss: 0.031833, Objective: 0.031833
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.40, NNZs: 1498, Bias: 0.000000, T: 2340, Avg. loss: 0.011833, Objective: 0.011833
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.89, NNZs: 1498, Bias: 0.000000, T: 2808, Avg. loss: 0.006963, Objective: 0.006963
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.70, NNZs: 1498, Bias: 0.000000, T: 3276, Avg. loss: 0.003606, Objective: 0.003606
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 3.23, NNZs: 1498, Bias: 0.000000, T: 3744, Avg. loss: 0.002387, Objective: 0.002387
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.26, NNZs: 1498, Bias: 0.000000, T: 4212, Avg. loss: 0.001090, Objective: 0.001090
```

```
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.37, NNZs: 1498, Bias: 0.000000, T: 4680, Avg. loss: 0.000979, Objective: 0.000979
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.13, NNZs: 1498, Bias: 0.000000, T: 5148, Avg. loss: 0.000429, Objective: 0.000429
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 2.36, NNZs: 1498, Bias: 0.000000, T: 5616, Avg. loss: 0.000378, Objective: 0.000378
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 2.36, NNZs: 1498, Bias: 0.000000, T: 6084, Avg. loss: 0.000210, Objective: 0.000210
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.25, NNZs: 1498, Bias: 0.000000, T: 6552, Avg. loss: 0.000183, Objective: 0.000183
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.75, NNZs: 1519, Bias: 0.000000, T: 468, Avg. loss: 0.770072, Objective: 0.770072
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.14, NNZs: 1519, Bias: 0.000000, T: 936, Avg. loss: 0.153449, Objective: 0.153449
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.72, NNZs: 1519, Bias: 0.000000, T: 1404, Avg. loss: 0.055433, Objective: 0.055433
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.37, NNZs: 1519, Bias: 0.000000, T: 1872, Avg. loss: 0.021970, Objective: 0.021970
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 2.73, NNZs: 1519, Bias: 0.000000, T: 2340, Avg. loss: 0.011175, Objective: 0.011175
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 2.42, NNZs: 1519, Bias: 0.000000, T: 2808, Avg. loss: 0.005127, Objective: 0.005127
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.36, NNZs: 1519, Bias: 0.000000, T: 3276, Avg. loss: 0.002768, Objective: 0.002768
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.59, NNZs: 1519, Bias: 0.000000, T: 3744, Avg. loss: 0.001614, Objective: 0.001614
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.70, NNZs: 1519, Bias: 0.000000, T: 4212, Avg. loss: 0.000761, Objective: 0.000761
```

```
00761
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 2.68, NNZs: 1519, Bias: 0.000000, T: 4680, Avg. loss: 0.000631, Objective: 0.000631
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.12, NNZs: 1519, Bias: 0.000000, T: 5148, Avg. loss: 0.000359, Objective: 0.000359
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.29, NNZs: 1519, Bias: 0.000000, T: 5616, Avg. loss: 0.000162, Objective: 0.000162
Total training time: 0.00 seconds.
-- Epoch 13
Norm: 1.22, NNZs: 1519, Bias: 0.000000, T: 6084, Avg. loss: 0.000100, Objective: 0.000100
Total training time: 0.00 seconds.
Convergence after 13 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.66, NNZs: 1536, Bias: 0.000000, T: 468, Avg. loss: 0.578926, Objective: 0.578926
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.36, NNZs: 1541, Bias: 0.000000, T: 936, Avg. loss: 0.074026, Objective: 0.074026
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.03, NNZs: 1543, Bias: 0.000000, T: 1404, Avg. loss: 0.018370, Objective: 0.018370
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.25, NNZs: 1547, Bias: 0.000000, T: 1872, Avg. loss: 0.006368, Objective: 0.006368
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.89, NNZs: 1547, Bias: 0.000000, T: 2340, Avg. loss: 0.002067, Objective: 0.002067
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.66, NNZs: 1547, Bias: 0.000000, T: 2808, Avg. loss: 0.000750, Objective: 0.000750
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.36, NNZs: 1547, Bias: 0.000000, T: 3276, Avg. loss: 0.000328, Objective: 0.000328
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.29, NNZs: 1547, Bias: 0.000000, T: 3744, Avg. loss: 0.000175, Objective: 0.000175
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.54, NNZs: 1547, Bias: 0.000000, T: 4212, Avg. loss: 0.000053, Objective: 0.000053
Total training time: 0.00 seconds.
-- Epoch 10
```

```
Norm: 1.22, NNZs: 1547, Bias: 0.000000, T: 4680, Avg. loss: 0.000027, Objective: 0.000027
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.26, NNZs: 1547, Bias: 0.000000, T: 5148, Avg. loss: 0.000009, Objective: 0.000009
Total training time: 0.00 seconds.
Convergence after 11 epochs took 0.00 seconds
-- Epoch 1
Norm: 1.44, NNZs: 1581, Bias: 0.000000, T: 468, Avg. loss: 0.738842, Objective: 0.738842
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 2.11, NNZs: 1586, Bias: 0.000000, T: 936, Avg. loss: 0.171131, Objective: 0.171131
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 1.56, NNZs: 1586, Bias: 0.000000, T: 1404, Avg. loss: 0.066483, Objective: 0.066483
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 1.97, NNZs: 1586, Bias: 0.000000, T: 1872, Avg. loss: 0.030961, Objective: 0.030961
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.28, NNZs: 1586, Bias: 0.000000, T: 2340, Avg. loss: 0.015704, Objective: 0.015704
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.85, NNZs: 1586, Bias: 0.000000, T: 2808, Avg. loss: 0.010254, Objective: 0.010254
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.69, NNZs: 1586, Bias: 0.000000, T: 3276, Avg. loss: 0.005178, Objective: 0.005178
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 1.11, NNZs: 1586, Bias: 0.000000, T: 3744, Avg. loss: 0.003250, Objective: 0.003250
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 1.75, NNZs: 1586, Bias: 0.000000, T: 4212, Avg. loss: 0.001426, Objective: 0.001426
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.21, NNZs: 1586, Bias: 0.000000, T: 4680, Avg. loss: 0.001263, Objective: 0.001263
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 2.94, NNZs: 1586, Bias: 0.000000, T: 5148, Avg. loss: 0.000654, Objective: 0.000654
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.28, NNZs: 1586, Bias: 0.000000, T: 5616, Avg. loss: 0.000554, Objective: 0.000554
Total training time: 0.00 seconds.
```

```
-- Epoch 13
Norm: 1.82, NNZs: 1586, Bias: 0.000000, T: 6084, Avg. loss: 0.000294, Objective: 0.000294
Total training time: 0.00 seconds.
-- Epoch 14
Norm: 1.66, NNZs: 1586, Bias: 0.000000, T: 6552, Avg. loss: 0.000244, Objective: 0.000244
Total training time: 0.00 seconds.
Convergence after 14 epochs took 0.00 seconds
-- Epoch 1
Norm: 0.96, NNZs: 1579, Bias: 0.000000, T: 468, Avg. loss: 0.775263, Objective: 0.775263
Total training time: 0.00 seconds.
-- Epoch 2
Norm: 1.72, NNZs: 1584, Bias: 0.000000, T: 936, Avg. loss: 0.162024, Objective: 0.162024
Total training time: 0.00 seconds.
-- Epoch 3
Norm: 2.34, NNZs: 1584, Bias: 0.000000, T: 1404, Avg. loss: 0.059058, Objective: 0.059058
Total training time: 0.00 seconds.
-- Epoch 4
Norm: 2.22, NNZs: 1586, Bias: 0.000000, T: 1872, Avg. loss: 0.029004, Objective: 0.029004
Total training time: 0.00 seconds.
-- Epoch 5
Norm: 1.72, NNZs: 1586, Bias: 0.000000, T: 2340, Avg. loss: 0.014697, Objective: 0.014697
Total training time: 0.00 seconds.
-- Epoch 6
Norm: 1.21, NNZs: 1586, Bias: 0.000000, T: 2808, Avg. loss: 0.007038, Objective: 0.007038
Total training time: 0.00 seconds.
-- Epoch 7
Norm: 1.94, NNZs: 1586, Bias: 0.000000, T: 3276, Avg. loss: 0.003723, Objective: 0.003723
Total training time: 0.00 seconds.
-- Epoch 8
Norm: 2.05, NNZs: 1586, Bias: 0.000000, T: 3744, Avg. loss: 0.002526, Objective: 0.002526
Total training time: 0.00 seconds.
-- Epoch 9
Norm: 2.48, NNZs: 1586, Bias: 0.000000, T: 4212, Avg. loss: 0.001530, Objective: 0.001530
Total training time: 0.00 seconds.
-- Epoch 10
Norm: 1.98, NNZs: 1586, Bias: 0.000000, T: 4680, Avg. loss: 0.001305, Objective: 0.001305
Total training time: 0.00 seconds.
-- Epoch 11
Norm: 1.59, NNZs: 1586, Bias: 0.000000, T: 5148, Avg. loss: 0.000800, Objective: 0.000800
Total training time: 0.00 seconds.
-- Epoch 12
Norm: 1.69, NNZs: 1586, Bias: 0.000000, T: 5616, Avg. loss: 0.000476, Objective: 0.000476
```


Out[72]:

▼ PassiveAggressiveClassifier ⓘ ?

► Parameters

C	1.0
fit_intercept	True
max_iter	1000
tol	0.001
early_stopping	False
validation_fraction	0.1
n_iter_no_change	5
shuffle	False
verbose	0
loss	'hinge'
n_jobs	None
random_state	42
warm_start	False
class_weight	None
average	False

```
In [73]: y_pred_best_pac_randomized = best_pac_classifier_randomized.predict(X_test_tfidf)
```

```
In [74]: accuracy_best_pac_randomized = accuracy_score(y_test, y_pred_best_pac_randomized)
classification_rep_best_pac_randomized = classification_report(y_test, y_pred_best_pac_randomized)
conf_matrix_test = confusion_matrix(y_test, y_pred_best_pac_randomized)
```

```
In [75]: print("Best PassiveAggressiveClassifier Model (RandomizedSearchCV):")
print(f"Best Hyperparameters: {best_params_randomized}")
print(f"Accuracy: {accuracy_best_pac_randomized}")
print("Classification Report:\n", classification_rep_best_pac_randomized)
```

Best PassiveAggressiveClassifier Model (RandomizedSearchCV):

Best Hyperparameters: {'verbose': 0, 'shuffle': False, 'fit_intercept': True, 'C': 1.0}

Accuracy: 0.7278911564625851

Classification Report:

	precision	recall	f1-score	support
negative	0.84	0.81	0.83	32
neutral	0.74	0.56	0.64	55
positive	0.68	0.83	0.75	60
accuracy			0.73	147
macro avg	0.75	0.74	0.74	147
weighted avg	0.73	0.73	0.72	147

```
In [76]: plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix_test, annot=True, fmt='d', cmap='Greys', xticklabels=['nega
plt.title('Confusion Matrix - Hyperparameters')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()
```

